LONDON METROPOLITAN UNIVERSITY

ITAHARI INTERNATIONAL COLLEGE

Module Code & Module Title

MA4001NT LOGIC AND PROBLEM SOLVING

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Year and Semester

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Student Name	London Met Id	College Id
Akisha Bhujel	24045800	np05cp4a240184
Manoj Katuwal	24045922	Np05cp4a240306
Biman hang Limbu	25045848	np05cp4a240069

Assignment Submission Date

MAY 20, 2025

Assignment Submitted To:

SANTOSH PARAJULI

I confirm that I understand my coursework needs to be submitted online via MySecondTeacher under the relevant module page before the deadline in order for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

ACKNOWLEDGEMENT

We deeply thank our module leaders, M.r Santosh Parajuli and M.r Romy Khatri for giving us a chance to solve these problems. Our team members Ms. Akisha Bhujel, Mr. Manoj Katwal and Mr. Biman Limbu, ideally completed the coursework. We couldn't have gotten any better teamwork and collaboration.

As a part of our assignment of module logic and problem solving we were asked to tackle three problems under limited time. We are very thankful for our group members who really worked hard and invested their precious time to complete this project. Each member gave their equal contribution and participated in problem solving and report writing. We could only complete this coursework with the active effort and cooperation of our team members.

A very special thanks to Itahari international college for giving us a chance to study this module and complete the coursework successfully. We are very grateful for college to provide us a diplomatic and friendly environment which helps to motivate us during difficult situations.

Overall, we are proud of our teammate's accomplishment. Their support and engagement in this project was everything. This experience was very beneficial in improving our logical thinking and problem solving skills.

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1 Introduction

This is a group coursework assigned by Itahari international college. Our group was formed randomly for the coursework, our group included Akisha Bhujel, Manoj katuwal and Biman Limbu. We collaborated on focusing on tackling three complex logical problems, using software tools like desmos graph, excel and google docs to complete our assigned task. These tools helped our team strengthen our skills and solve challenges effectively. Our team collaboration was the pivot of our strength and expertise to complete logical problems.

This coursework gave us a lot of new experience for solving tough logical problems and gaining knowledge on new concepts of mathematical problems. We are very grateful to everyone who guided us when we couldn't solve it and helped us tackle these problems throughout the process.

We want to start by expressing our heartfelt gratitude towards our teachers and classmates at Itahari international college for their sincere and helpful advice. Their support has been a crucial role in helping us complete our assignment with quality and submitted within the deadline. our especial thanks to Mr Romy Khatri for guiding us during our coursework.

A huge thanks to our mentors and classmates at Itahari international college for their super helpful and kind nature. Their support was essential in helping us finish our coursework within the given time period.

Working with such dedicated members has been a joy to me. Their support has been the pillar of my coursework and I am especially grateful for their precious time and efforts.

Problem 1

Write a Procedure, tax to calculate (in Nepalese Rupees) the tax a person owes, depending on his/her income. Calculate the tax using this table.

Taxable income (Rs)	Income tax rates (Percent)
0 to 50,000	1%
50,000 to 2,00,000	10%
2,00,000 to 3,00,000	20%
3,00,000 to 10,00,000	30%
10,00,000 to 30,00,000	36%
30,00,000 and above	39%

Table 1:Tax Slab

Showing the following:

I. The salary.
II. The tax rate.
III. The amount of tax.
IV. The amount left after tax and.
V. Be able to deal with any input valid or not.

Our test of procedure include the following values, which should be included in our report.

a) Tax (Rs 4,05,000)

Deduction:

- Employee provided fund organization (Rs 10,000)
- Life insurance premium (Rs 68000)
- Citizen Investment Trust (Rs 0)

b) Tax (Rs 8,09,090)

Deduction:

- Employee provided fund organization (Rs 15000)
- Life insurance premium (Rs 20,000)
- Citizen Investment Trust (Rs 5,000)

c) Tax (Rs 19,10,000)

Deduction:

- Employee provident fund organization (Rs 0)
- Life insurance premium (Rs 2,50,000)
- Citizen Investment Trust (Rs 50,000)

d) Tax (Rs 2,108,790)

Deduction:

- Employee provident fund organization (Rs 1,50,000)
- Life insurance premium (Rs 30,000)
- Citizen Investment Trust (Rs 25,000)

e) Tax (Rs 31,800)

Deduction:

- Employee provident fund organization (Rs 0)
- Life insurance premium (Rs 5,000)
- Citizen Investment Trust (Rs 0)

f) Tax (Rs -50,000)

Deduction:

- Employee provident fund organization (Rs 0)
- Life insurance premium (Rs 0)
- Citizen investment trust (Rs 0)

1.1 Tax

Tax is a mandatory charge of money that individuals and businessman must pay to the government. The taxes are used for the services of the citizens by improving or building public infrastructures such as school, roads, bridge, playground, transportation etc. People can pay the tax in Nepal through both online and offline methods. In Nepal, there are different types of tax like income tax, sales tax and property tax. Paying tax helps the country to develop and function more properly.

1.1.1 Different type of tax in Nepal

- Income tax: Income tax is a tax paid by the individual and businessman based on the money amount how much they earn. In Nepal, people with certain level of income must has to paid the tax to the government.
- Value Added Tax: Value Added Tax is a type of tax charged on the sale of goods and services. When you buy some goods some amount of money goes to the government as VAT tax. In Nepal, the VAT rate is set to be 13%. This type of tax is collected by seller from the customer and paid to the government.
- Property tax: Property tax is a type of tax paid by the individual or the organization who own the house, land or building. In Nepal, this type of tax is collected by the local government or municipality for the development of local infrastructure.

(thapa, 2025)

1.1.2 Importance of tax

Tax is very important for the development and functionality of the country. It is a main source of the government where government can provide public services in education, health, educator sector. In the context of Nepal, it is very challenging to provide these facilities without tax. Lastly for the well-being of the citizen the tax is very important.
(thapa, 2025)

1.2 Tax Calculation Format

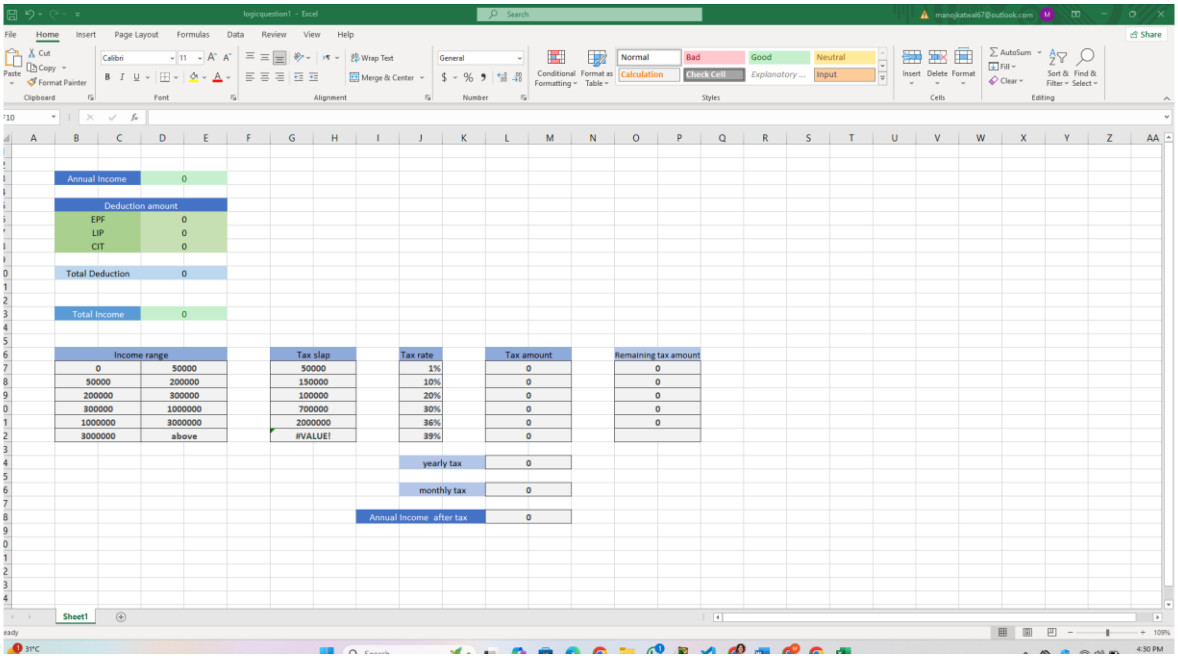


Figure 1: Tax Calculation Format

1.3 Tax Calculation Formula

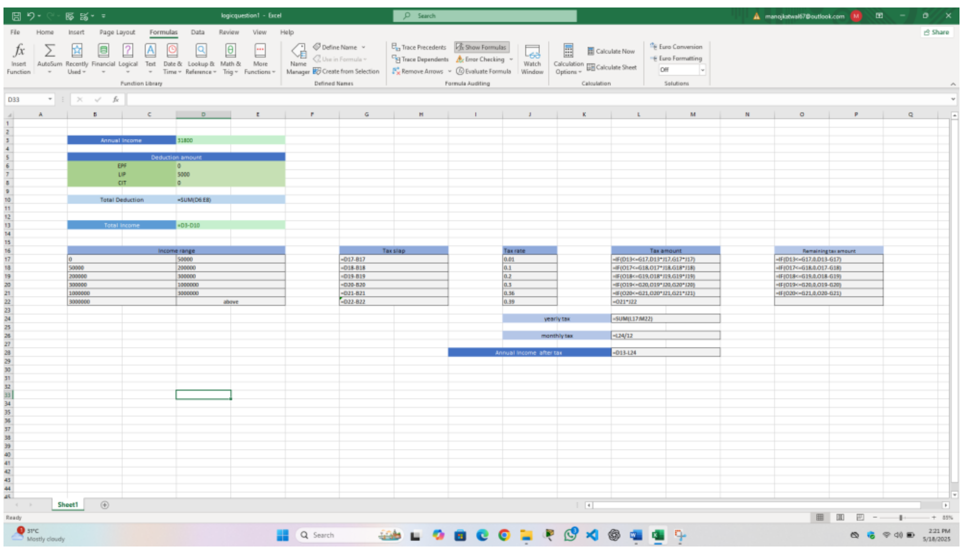


Figure 2:Tax Calculation Formula

1.4 Tax calculation of different income with different education

1.4.1 Tax (Rs 4,05,000)

Deduction:

- Employee provided fund organization (Rs 10,000)
- Life insurance premium (Rs 68000)
- Citizen Investment Trust (Rs 0

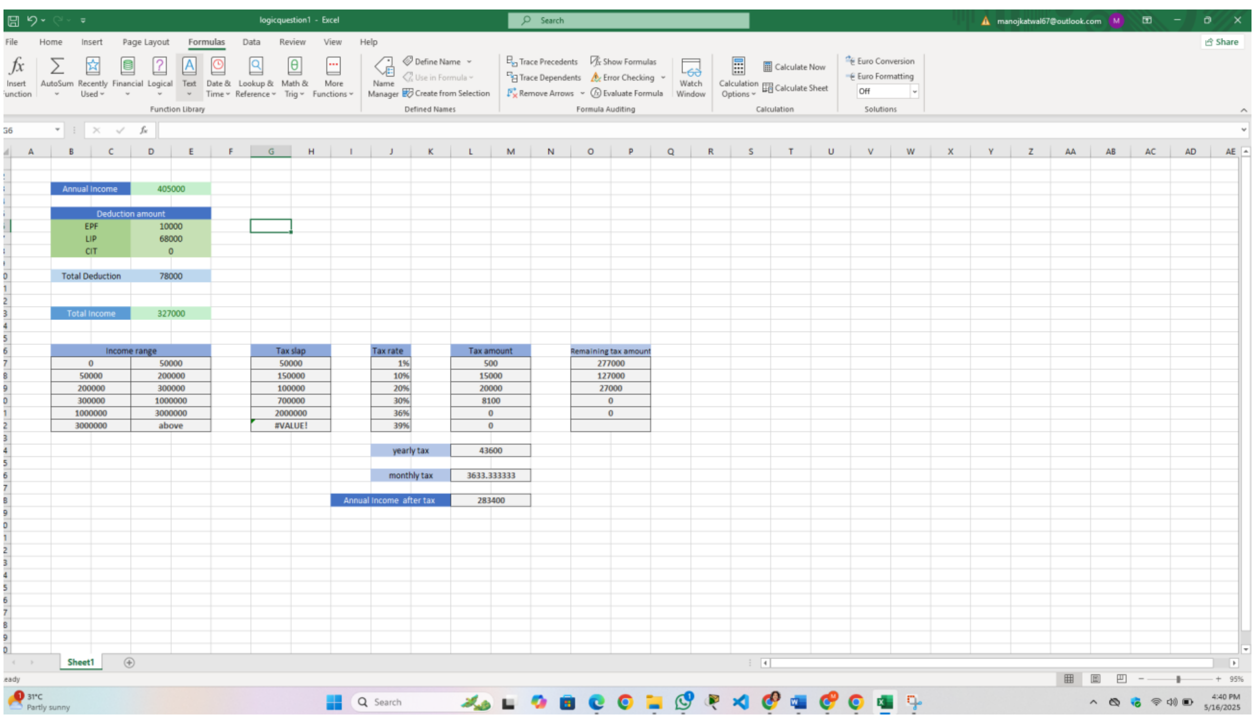


Figure 3:Test 1 calculation

If we take 4,05,000 as the annual income for the tax calculation after some deduction like in employee provident fund Rs 10,000 and in life insurance premium Rs 68000. Them we have to paid yearly tax Rs 4360 and monthly ta be Rs 3633 and total annual income after tax became Rs 283400.

1.4.2 Tax (Rs 8,09,090)

Deduction:

- Employee provided fund organization (Rs 15000)
- Life insurance premium (Rs 20,000)
- Citizen Investment Trust (Rs 5,000)

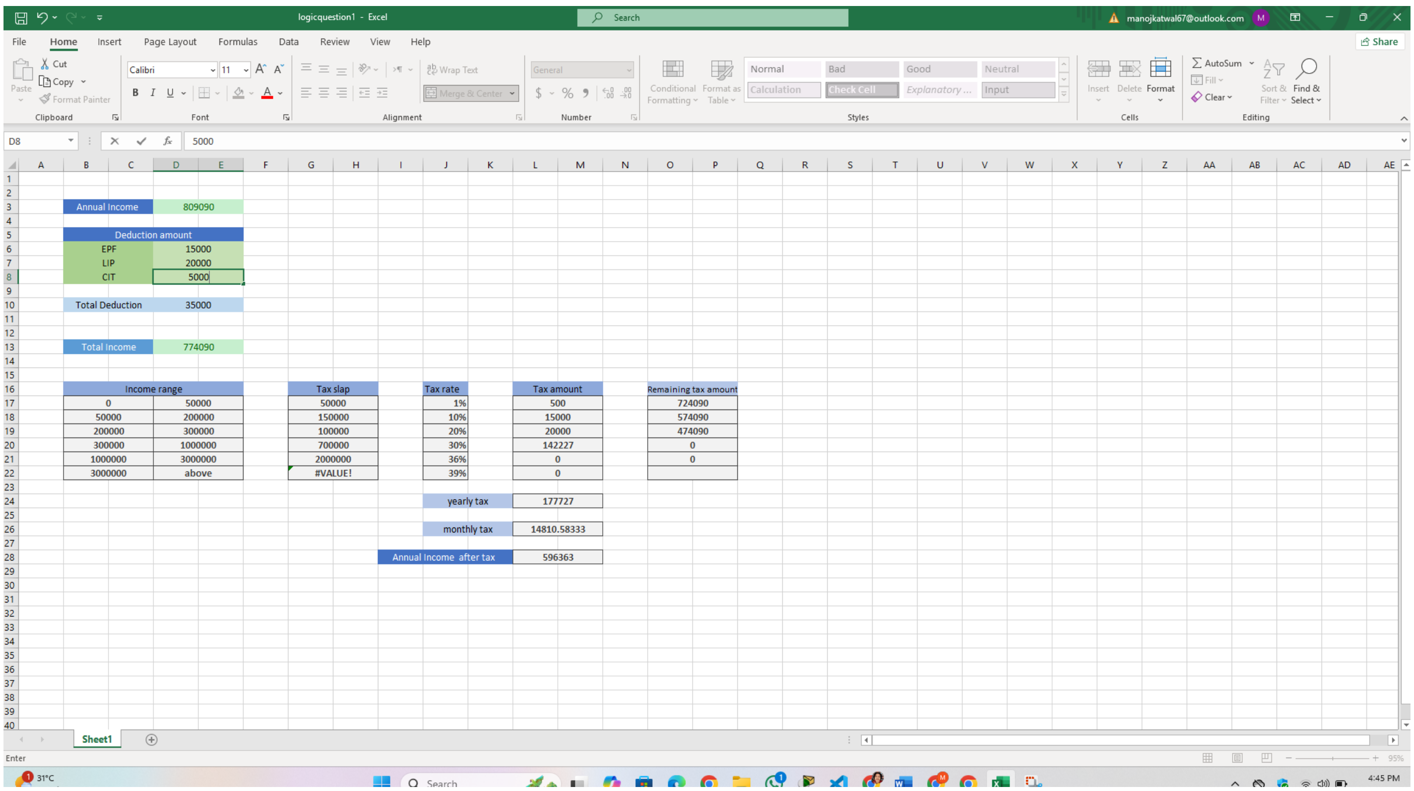


Figure 4:Test 2 Calculation

If we take 8,09,090 as the annual income for the tax calculation after some deduction like in employee provident fund Rs 15,000 and in life insurance premium Rs 20,000 and in citizen investment trust Rs 5000. Then we have to paid yearly tax Rs 177727 and monthly ta be Rs 14810 and total annual income after tax became Rs 596363.

1.4.3 Tax (Rs 19,10,000)

Deduction:

- Employee provident fund organization (Rs 0)
- Life insurance premium (Rs 2,50,000)
- Citizen Investment Trust (Rs 50,000)

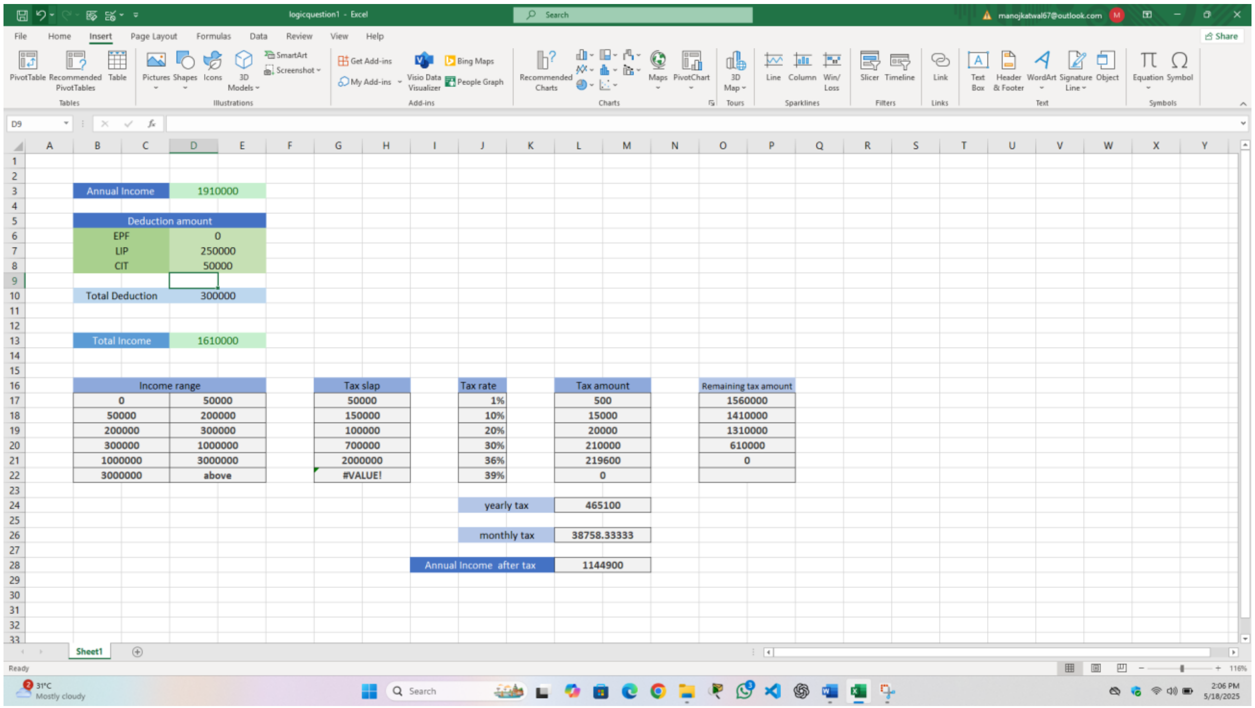


Figure 5:Test 3 Calculation

If we take 1910000 as the annual income for the tax calculation after some deduction like in life insurance premium Rs 25,000 and in citizen investment trust Rs 50000. Then we have to paid yearly tax Rs 481300 and monthly ta be Rs 40108 and total annual income after tax became Rs 1173700.

1.4.4 Tax (Rs 2108790)

Deduction:

- Employee provident fund organization (Rs 1,50,000)
- Life insurance premium (Rs 30,000)
- Citizen Investment Trust (Rs 25,000)

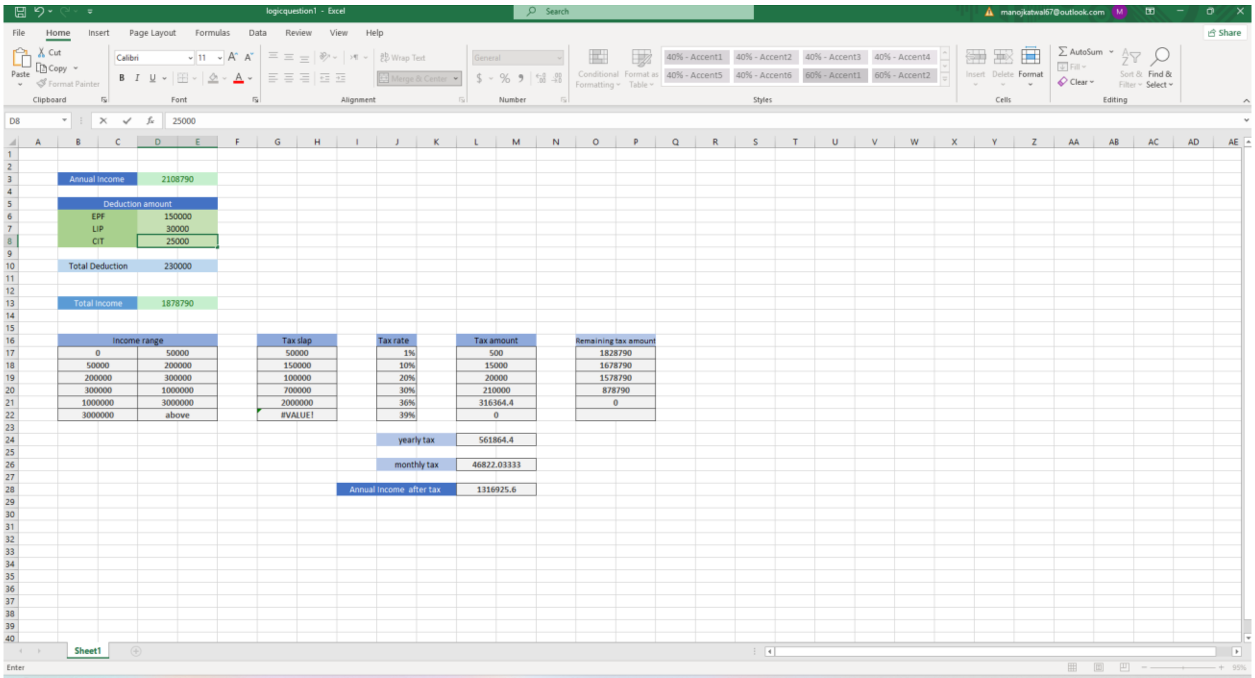


Figure 6:Test 4 Calculation

If we take 2108790 as the annual income for the tax calculation after some deduction like in employee provident fund Rs 1,50,000 and in life insurance premium Rs 30,000 and in citizen investment trust Rs 25000. Then we have to paid yearly tax Rs 561864 and monthly ta be Rs 46822 and total annual income after tax became Rs 1316925.

1.4.5 Tax (Rs 31800)

Deduction:

- Employee provident fund organization (Rs 0)
- Life insurance premium (Rs 5,000)
- Citizen Investment Trust (Rs 0)

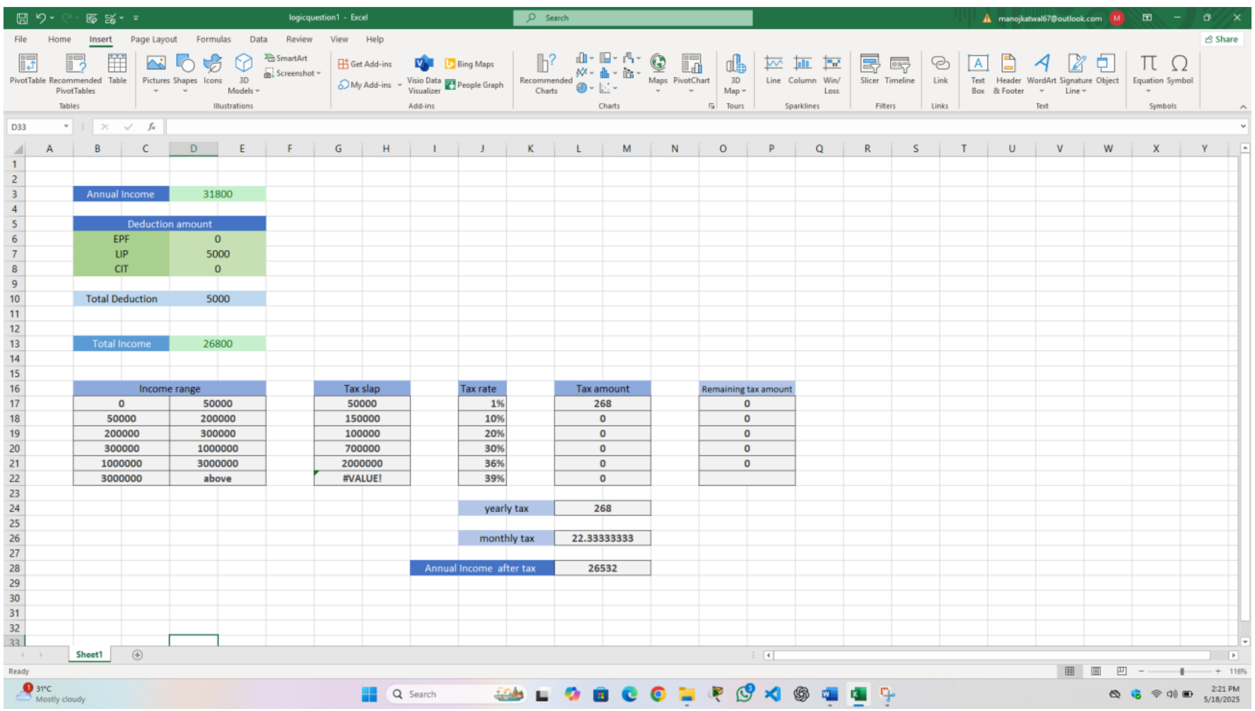


Figure 7:Test 5 tax calculation

If we take 31800 as the annual income for the tax calculation after some deduction like in life insurance premium Rs 5000. Then we have to paid yearly tax Rs 18 and monthly ta be Rs 1.5and total annual income after tax became Rs 1782.

1.4.6 Tax (Rs -50,000)

Deduction:

- Employee provident fund organization (Rs 0)
- Life insurance premium (Rs 0)
- Citizen investment trust (Rs 0)

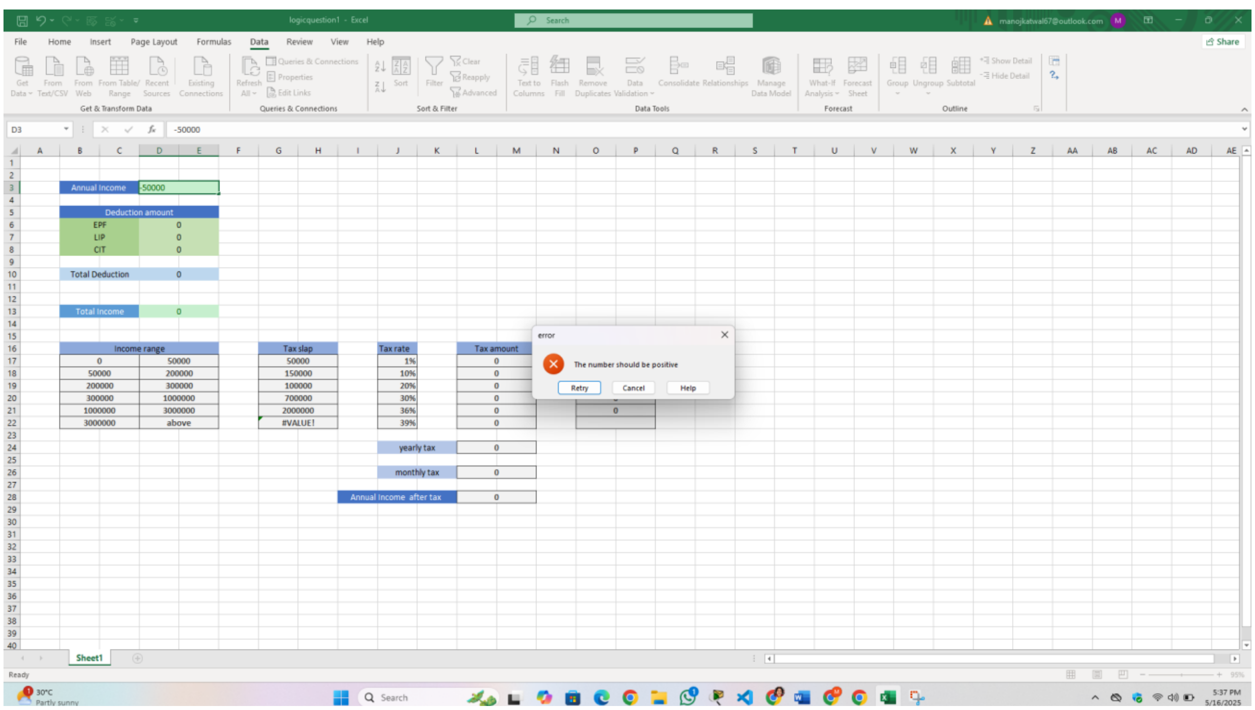


Figure 8: Test 5 tax calculation

If we take -5000 as the annual income for the tax calculation. The system gives a pop up error message.

2 Problem 2

2.1 Part A

Berkshire Health Care patient diet is to contain at least 8 units of vitamins, 9 units of minerals, and 10 carbohydrates. Three foods, Food A, Food B, and Food C are to purchased. Each unit of Food A provides 1 unit of vitamins, 1 unit of minerals, and 2 carbohydrates. Each unit of Food B provides 2 units of v

Berkshire Health Care patient diet is to contain at least 8 units of vitamins, 9 units of minerals, and 10 carbohydrates. Three foods, Food A, Food B, and Food C are to be purchased. Each unit of Food A provides 1 unit of vitamins, 1 unit of minerals, and 2 carbohydrates. Each unit of Food B provides 2 units of vitamins, 1 unit of minerals and 1 carbonhydrates .Each unit of food C provides 2 units of vitamin 1,unit of minerals and 2 carbonhydrates. If Food A costs $3 per unit, Food B costs $2 per unit and Food C costs $3 per unit, how many units of each food should be purchased to keep costs at minimum.

Questions

You should answer the following questions and incorporate your answers into a word-processed report to form part of your final pdf. The sections of your report should correspond to the individual questions following.

a) Formulate the problem as a linear programming model, clearly defining the variables, the objective function, and the constraints.

b) Solve the problem using the Simplex method.

c) Solve the problem using the Excel Solver and interpret the results.

d)For the final part of your report, in your capacity as an Adviser, you should present a memorandum to Berkshire Health Care. Describe your main conclusions in simple, non-technical English, i.e., do not use technical terms like variable, objective function, or dual price. Don’t worry about repeating some or all the points that you have already made in answer to earlier questions. The aim is to communicate your conclusions clearly to someone who is knowledgeable about the combination of food used in the diet plan, but who knows nothing about the subject of linear programming. You may use tables and charts if you wish.

a) Problem Analysis

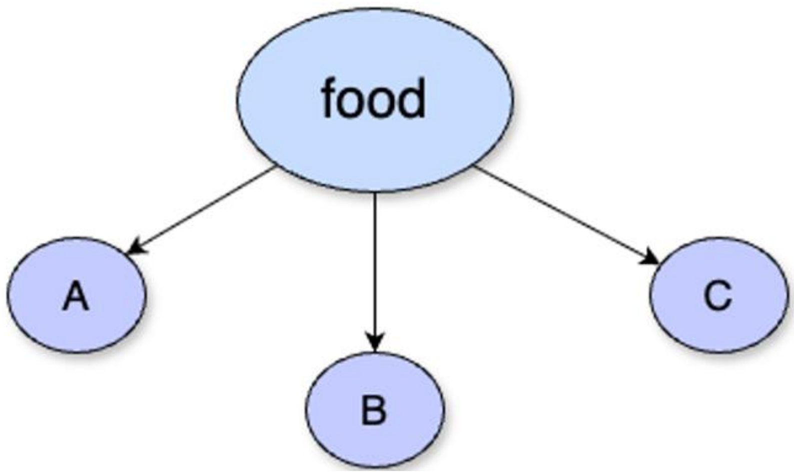


Figure 9: Problem Analysis of part A

Proteins	Food A	Food B	Food C	requirement
vitamins	1	2	2	8
minerals	1	1	1	9
carbohydrates	2	1	2	10
cost	3	2	3	-

Table 2:Table of problem analysis of part A

Decision variable

Here, there will be three decision variables x, y and z to represent the quantity of food A, food B and food C to minimize the cost and protein requirement.

Objective function

Since the cost of three food items are given we have to minimize the cost given.

Minimize Z' = 3x + 2y +

Constraints

x + 2y + 2z >= 8 (for vitamins)

x + y + z >= 9 (for minerals)

2x + y + 2z >= 10 (for carbohydrates)

x, y, z >= 0 (non-negative constraints)

b) **Using simplex method**

Let -S1, -S2 and -S3 be surplus and A1, A2 and A3 be artificial value

x + 2y + 2z - S1 + A1 = 8

x + y + z - S2 + A2 = 9

2x + y + 2z - S3 + A3 = 10

x, y, z, S1, S2, S3, A1, A2, A3 >= 0

Standard equation

1z' - 3x - 2y - 3z + 0S1 + 0S2 + 0S3 - 10A1 - 10A2 - 10A3 = 0

0z' + 1x + 2y + 2z - S1 + 0S2 + 0S3 + A1 + 0A2 +0 A3 = 8

0z' + x + y + z + 0S1 - S2 + 0S3 + 0A1 + A2 + 0A3 = 9

0z' + 2x + y + 2z + 0S1 + 0S2 - S3 + 0A1 + 0A2 + A3 = 10

z', x, y, z S1, S2, S3, A1, A2, A3, >= 0

Table No 1

We have to make artificial value 0 before continuing the solution.

Rows	z'	x	y	z	S1	S2	S3	A1	A2	A3	Constant
R0	1	-3	-2	-3	0	0	0	-10	-10	-10	0
R1	0	1	2	2	-1	0	0	1	0	0	8
R2	0	1	1	1	0	-1	0	0	1	0	9
R3	0	2	1	2	0	0	-1	0	0	1	10

Table 3:Initial table of part A

Table no 2

For new identity matrix,

New R0 = old R0 + 10(R1 + R2 + R3)

Rows	z’	x	y	z	S1	S2	S3	A1	A2	A3	Constant
R0	1	37	38	47	-10	-10	-10	0	0	0	0
R1	0	1	2	2	-1	0	0	1	0	0	8
R2	0	1	1	1	0	-1	0	0	1	0	9
R3	0	2	1	2	0	0	-1	0	0	1	10

Table 4:Table 2 of part A

After making the value of artificial A1, A2 and A3 to zero. Here, the highest positive number is 47 and the smallest ratio is in R1 so, 2 is our pivot element

New R1 = old R1/ pivot element

New R0 = old R0 - 47 X new R1

New R2 = old R2 - new R1

New R3 = old R3 - 2 X new R2

Table 3

Rows	z’	x	y	z	S1	S2	S3	A1	A2	A3	Constant
R0	1	27/2	-9	0	27/2	-10	-10	-47/2	0	0	82
R1	0	1/2	1	1	1/2	-1	0	-1/2	0	0	4
R2	0	1/2	0	0	1/2	-1	0	-1/2	1	0	5
R3	0	1	-1	0	1	0	-1	-1	0	1	2

Table 5:Table 3 of part A

Here, 27/2 is the highest positive number and 2 is lowest ratio so, 1 is pivot

New R0 = old R0 - 27/2 X new R3

New R1 = old R1 - 1/2 X new R3

New R2 = old R2 - 1/2 X new R3

Table 4

Rows	z’	x	y	z	S1	S2	S3	A1	A2	A3	Constant
R0	1	0	9/2	0	0	-10	7/2	-10	0	-27/2	55
R1	0	0	3/2	1	-1	0	1/2	1	0	-1/2	3
R2	0	0	1/2	0	0	-1	1/2	0	1	-1/2	4
R3	0	1	-1	0	1	0	-1	-1	0	1	2

Table 6:Table 4 of part A

From the table -5, 9/2 is the highest value and 2 is the lower ratio so, 3/2 is key element

New R1 = old R1/key element

New R0 = old R0 - 9/2 X new R1

New R2 = olde R2 - 1/2 X new R1

New R3 = old R3 + new R1

Table 5

Rows	z’	x	y	z	S1	S2	S3	A1	A2	A3	Constant
R0	1	0	0	-3	3	-10	2	-13	0	-12	46
R1	0	0	1	2/3	-2/3	0	1/3	2/3	0	-1/3	2
R2	0	0	0	-1/3	1/3	-1	1/3	-1/3	1	-1/3	3

R3	0	1	0	2/3	1/3	0	-2/3	-1/3	0	2/3	4

Table 7:Table 5 of part A

From the table - 5, highest value is 3 and lowest positive value is 3 and lowest positive value is 3 so, key element is 1/3 key element.

New R2 = old R2/key element

New R0 = old R0 - 3 X new R2

New R1 = old R1 + 2/3 X new R2

New R2 = old R3 - 1/3 X new R2

Table 6

Rows	z'	x	y	z	S1	S2	S3	A1	A2	A3	Constant
R0	1	0	0	0	0	-7	5	-22	0	-9	19
R1	0	0	1	0	0	1/3	1	0	2	-7/6	8
R2	0	0	0	-1	1	-3	1	-1	3	-1	9
R3	0	1	0	1	0	1	-1/3	0	-1	1	1

Table 8:Table 6 of part A

Here, all the coefficient values in the R0 row are <= 0 so, it reached the optimum solution.

Hence,

Minimum Z' = 19

When x = 1, y = 8 and z = 0

c) Solving the problem using Excel Solver

At first, Initializing the value in excel

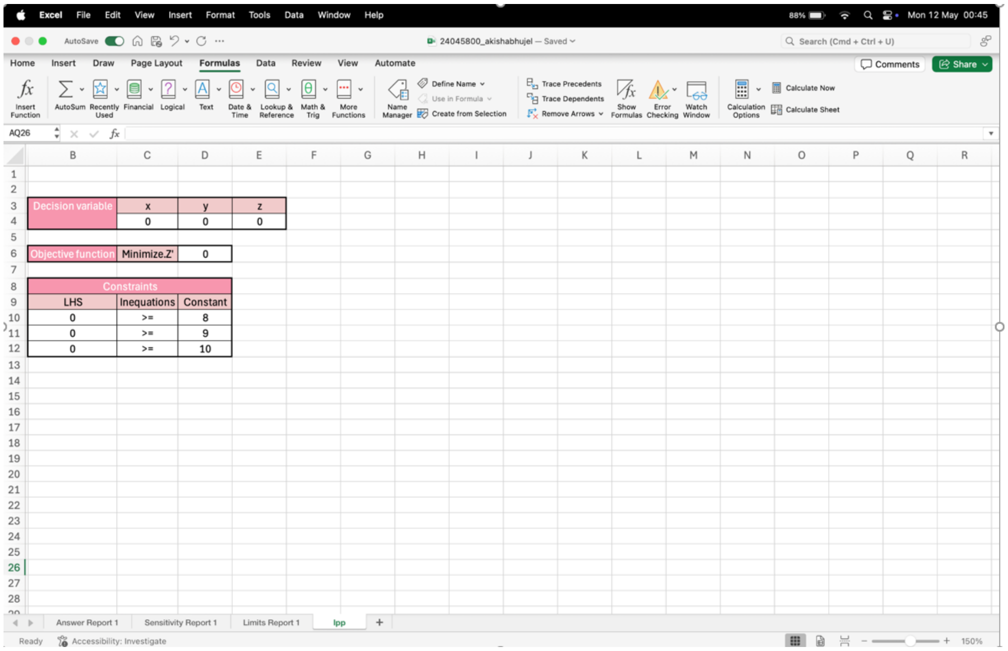


Figure 10: Initial value of excel

Implementing the formula

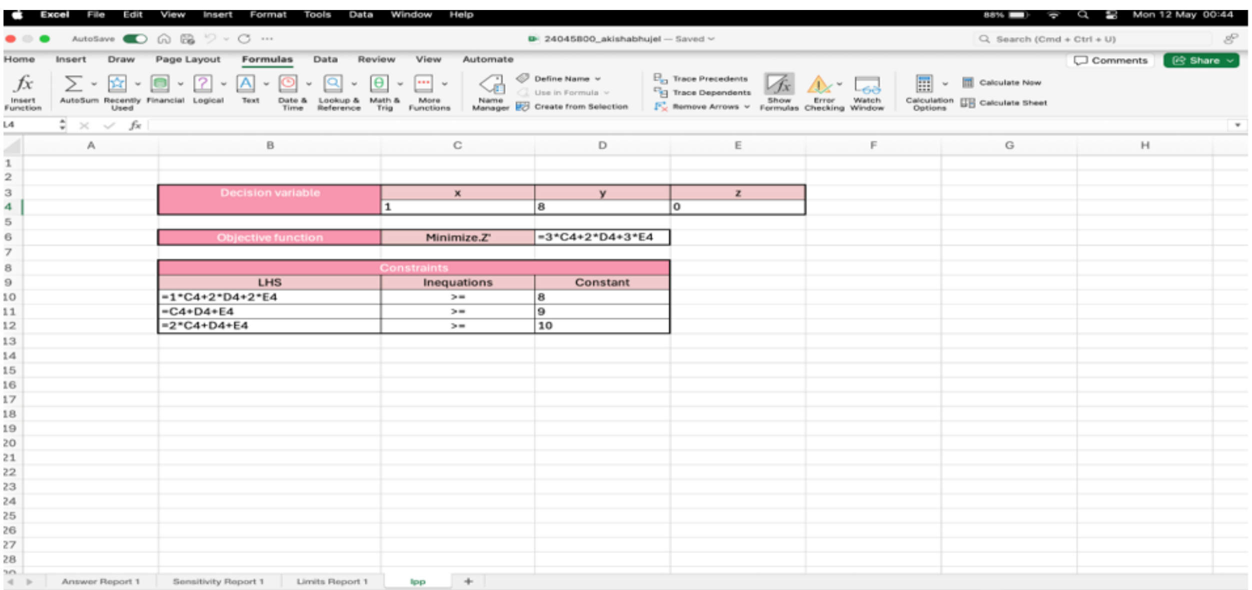


Figure 11:Inserting the formula

Solving the problem using Excel Solver

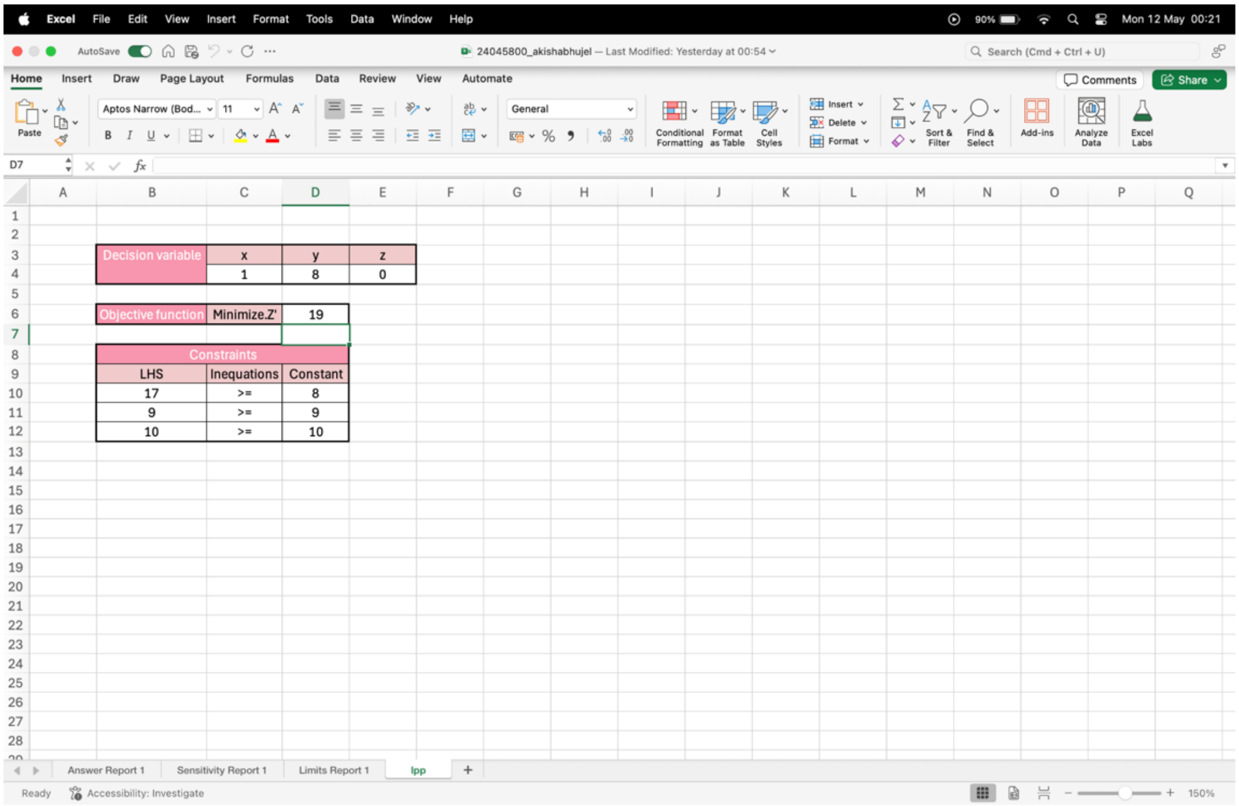


Figure 12:Solving the problem in excel

Answer report:

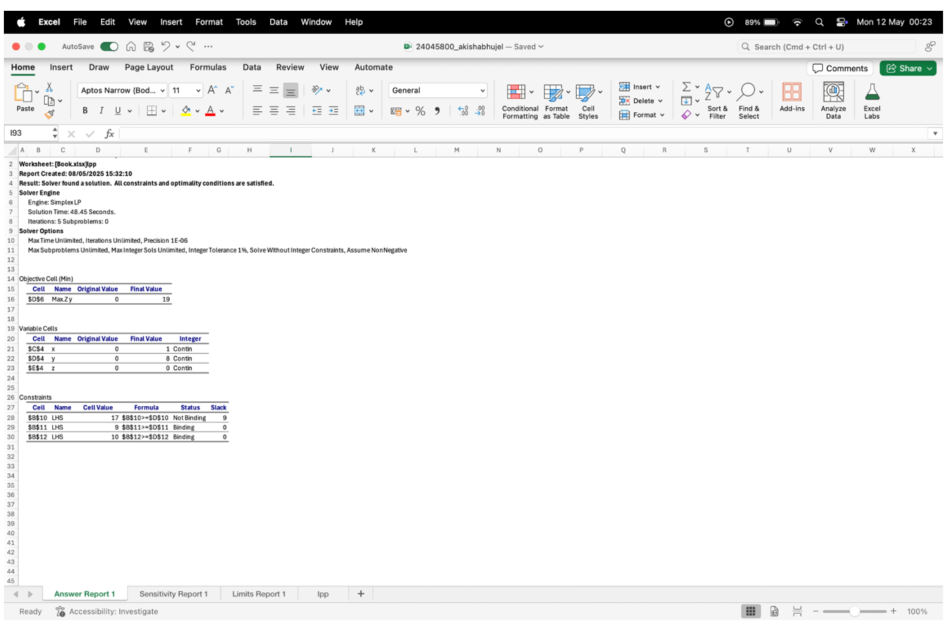


Figure 13:Answer report

Sensitivity Report:

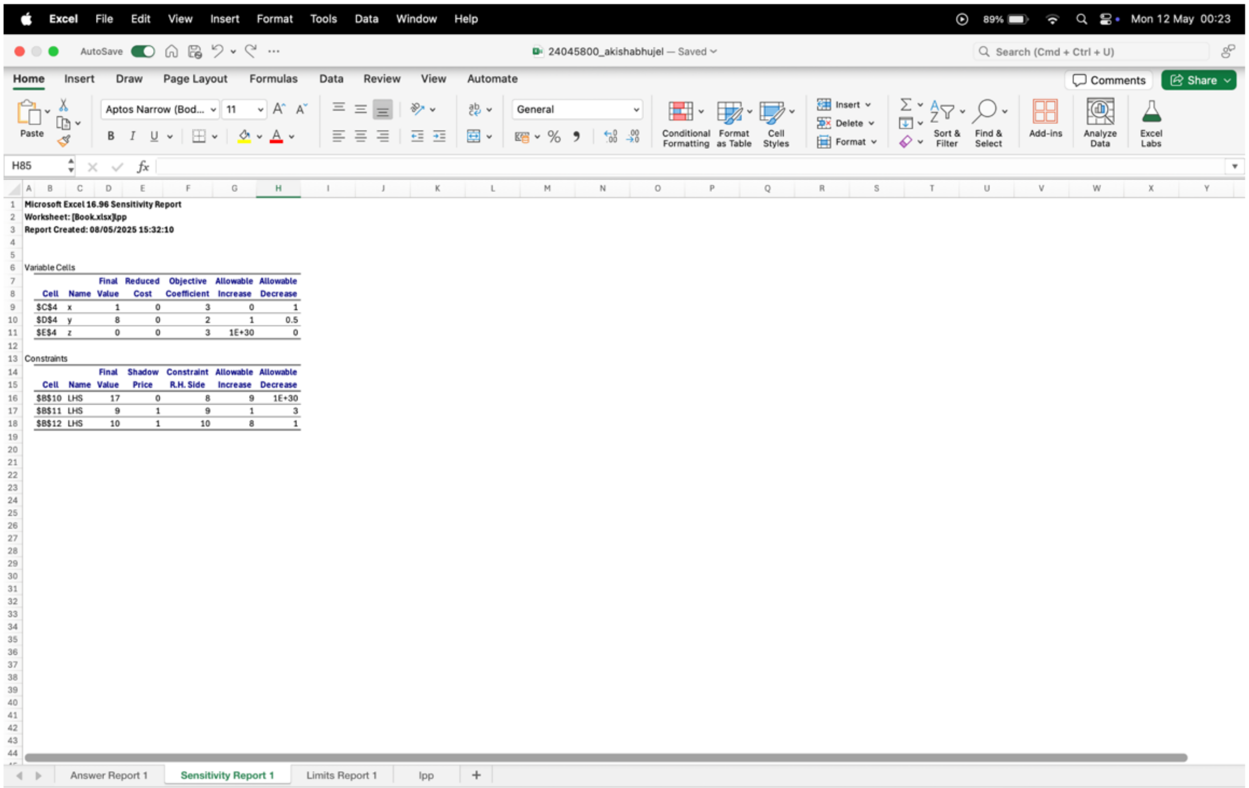


Figure 14:Sensitivity Report

Limits Report:

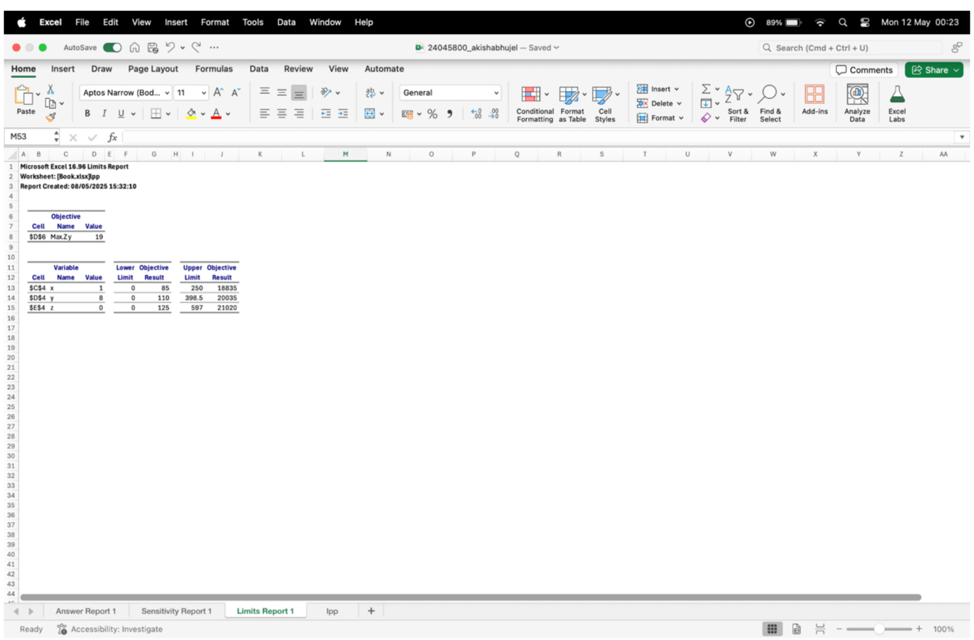


Figure 15:Limits Report

d)

Date : May 11, 2025

To : Berkshire Health Care Organization

From : Akisha Bhujel

Subject : Recommendation of nutritional diet for patients

Dear Berkshire Health Team,

I have detailed diet goals for the patient to ensure their nutritional requirement is reached with the lowest cost, the best combination of food with highest nutritional content and cheap cost. The diet must supply 8 units of vitamins, 9 units of minerals and 10 units of carbohydrates serving three food items A, B and C.

After analyzing the protein source provided by each food and cost implications, I have acknowledged the best source of vitamins, minerals and carbohydrates for patients are 8 units of vitamins, 9 units of minerals and 10 units of carbohydrates. This combination of food gives the patient the minimum required amount of nutritions. I have summarized my view under a table to give you a better understanding.

Table to display food items and their nutritional value

Nutritions	Food A	Food B	Food C	Benefits
Vitamins	1	2	2	17
Minerals	1	1	1	9
Carbohydrates	2	1	2	10

As displayed on the table the food items do meet the minimum requirement of the protein addressed, with the minimal cost. After calculation I have concluded that the cost for the minimum unit of protein is 19. If food A and food B are given in units of 1 and 8 then we can fulfill our protein intake at a minimum cost.

Thank you for your consideration. I assure you that this will benefit your team and meet targeted goals. Please feel free to contact me if you are planning to put it in action or would like me to add more points. I am delighted to assist.

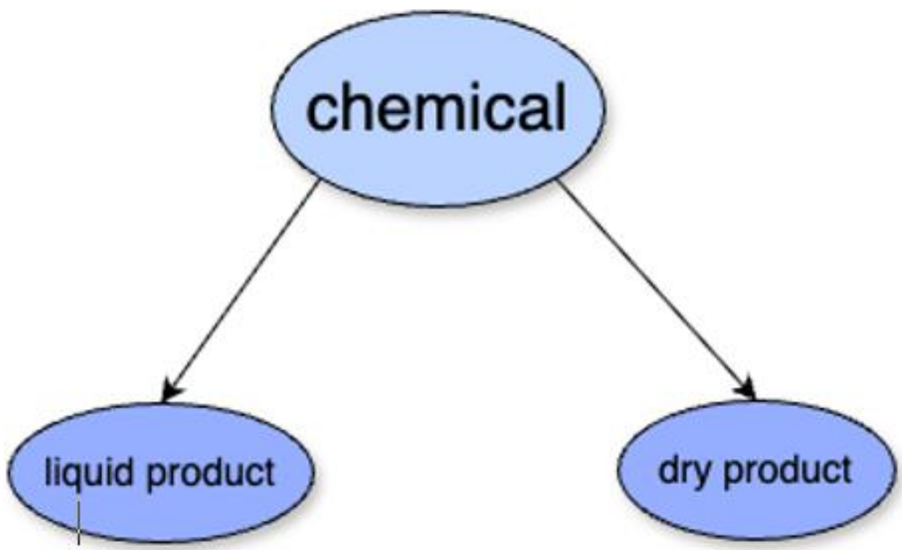
Sincerely,

Akisha Bhujel

2.2 Part B

Mr. Harris requires a minimum of 10,12 and 12 units of chemicals A, B and C respectively for his garden. A liquid product contains 5, 2 and 1 units of A, B and C respectively per jar. A dry product contains 1, 2 and 4 units of A, B and C per carton. If the liquid product cost £ 3 per jar and dry product cost £ 2 per carton, how many of each should Harris purchase in order to minimize the cost and meet the requirements. **Formulate this problem and solve it graphically.**

a) Problem analysis



Product	**Liquid**	**Dry**	**Constraint**
Chem A	5	1	10
Chem B	2	2	12
Chem C	1	4	12
cost	3	2	-

Table 9: Table of data analysis part B

Decision variable

Let x and y be the number of units of liquid and dry product that should be produced in order to minimize the cost and meet requirements.

Objective function

Minimize Z = 3x + 2y

Constraints

5x + 1y >= 10 (chem A)

2x + 2y >= 12 (chem B)

1x + 4y >= 12 (chem C)

x, y >= 0 (non-negative constraints)

Graphical Solution

Changing above inequalities to equalities. We get,

5x + 1y >= 10 ——————(i)

2x + 2y >= 12 ——————(ii)

1x + 4y >= 12 ——————(iii)

From equation 1

5x + 1y >= 10

x	0	10
y	2	0

Equation (i) passes through (0,10) and (2,0)

Origin test:

Put x = 0 and y = 0 in 5x + 1y >= 10,

0 > 10 which is false, so the origin lies outside the equation.

From equation 2

2x + 2y >= 12

x	0	6
y	6	0

Equation (ii) passes through (0,6) and (6,0)

Origin test:

Put x = 0 and y = 0 in 2x + 2y >= 12,

0 >= 12 which is false, so the origin lies outside the equation.

From equation 3

1x + 4y >= 12

x	0	3
y	12	0

Equation (iii) passes through (0,3) and (12,0)

Origin test:

Put x = 0 and y = 0 in 1x + 4y >= 12,

0 >= 12 which is false, so the origin lies outside the equation.

From the graph

vertex	**x**	**y**	**3x + 2y**
(1,5)	1	5	13
(4,2)	4	2	16

Hence, the minimum value of Z will be 13 when, x = 1 and y = 5 therefore the chemical contains 1 unit of liquid product and 5 units of dry product to minimize cost of 13.

Graphical representation

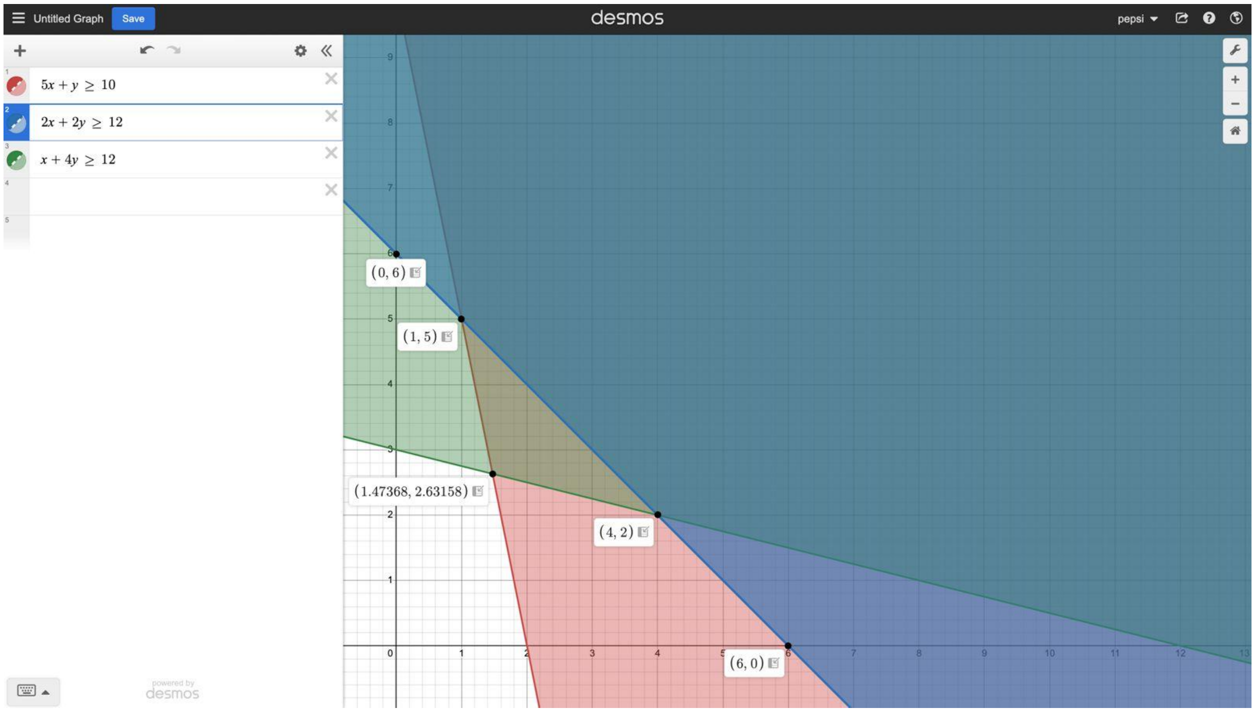


Figure 16:Graph of part B

3 Problem 3

If the revenue function for a manufacturing company is R(x) = 10 – x and the total cost function is C(x) = x2 - 8x + 20, where x is output.

You should answer the following questions and incorporate your answers into a word-processed report. The sections of your report should correspond to the individual question below.

Questions

a) Find the breakeven points (round in 3 decimal places).

Briefly explain the breakeven points in terms of number of items and cost.

b) Plot the revenue and cost functions on graph, showing the accurate break- even output levels and corresponding price.

c) Determine the profit function $P(x)$, for the company and find,

(i) The level production that maximizes the profit.

(ii) The maximum profit.

Solution,

Given,

• Revenue Function (R(x) = 10 -x

Cost Function (C(x) = x^2-8x+20

At Break even point,

(R(x) | = (c(x))

Or, 10-x = x^2-8x+20

Or, 0 = x^2-8x-x+20

Or, 0 = x^2-7x+10

X^2-7x+10=0

No,

Comparing with quadratic equation.

ax^2+10x+C= 0 we got

a=7 b=7 C=10

P.T.O

$$x = \frac{7 \pm \sqrt{(-7)^2 - 4(1)(10)}}{2(1)}$$

$$= \frac{7 \pm \sqrt{49-40}}{2}$$

$$= \frac{7 \pm \sqrt{9}}{2}$$

Taking +ve,

$$= \frac{7 \pm \sqrt{9}}{2}$$

$$= \frac{10}{2}$$

$=5$

Taking -ve,

$$= \frac{7 - \sqrt{9}}{2}$$

$$= \frac{4}{2}$$

$=2$

The break-even points are x=5 and x=2.

The breakeven points occurs where the firm produces an output such that it recovers the total revenue Collected and its total Costs.

Especially

- If the quality is 2 items, it neither loss or gain as it covers its costs.
- With the 5 items as well, things are similar revenue paid for selling 5 items also makes up the costs.

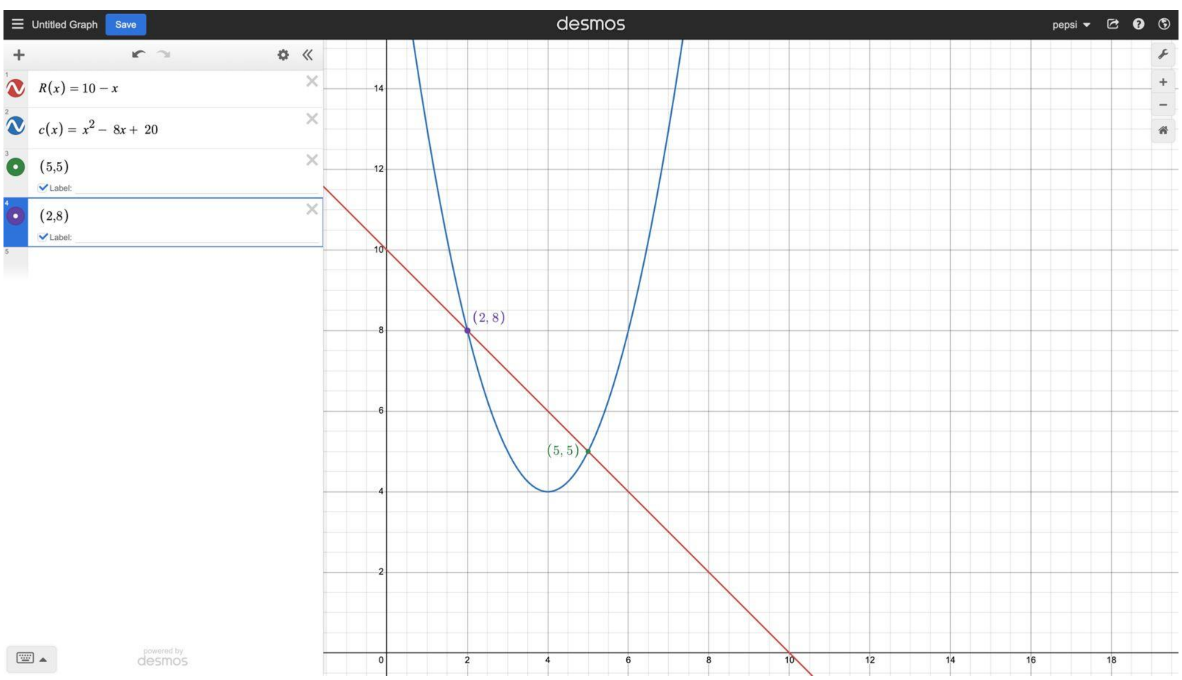


Figure 17:Problem 3 output x

Here, the graph showing the revenue and cost functions, with the break-even points clearly at (2, 8) and (5, 5).

i) Profit function,

P(x) = R(x) - C(x)

=(10-x) – (x^2-8x+20)

=10-x-x^2+8x-20

=-x^2=7x-10

x = $\frac{-b}{2a}$

Where a=-1 and b =7

x = $\frac{7}{2(-1)}$

=$\frac{7}{2}$

ii) Maximum Profit,

= $\frac{c-b}{4a}$ c-b^2

=-10- $\frac{(7)}{4*-1}$

=$\frac{-10*-4-49}{-4}$

=$\frac{-9}{-4}$

= $2.25

Therefore, the level production that minimizes the profit is 3.5 units and maximum p ofits is $2.25.

4 Conclusion

This coursework provides us a opportunity to get knowledge about logical and problem-solving skill in real-world context such as in tax calculation, diet planning, using linear programming problems and break-even analysis. Our team has divided the task equally which helps us to do this coursework more faster and easier. While doing coursework we faced several problems in calculating the tax correctly with different salary income and certain deduction from the income. Using Excel Solver for solving problems became a bit difficult at first but we managed this challenges by dividing task and discussing the ideas in the meeting.

Lastly our hard work and team work helps us to successful to solve the problem. This coursework helps us to explore important ideas related to business analysis and decision making. Through this coursework we successfully managed to developed critical thinking , team work, and gain the knowledge regarding the tax calculation, break-even points etc.

5 References

thapa, P., 2025. *an-overview-of-tax-law-in-nepal/.* [Online]
Available at: https://primelawnepal.com/an-overview-of-tax-law-in-nepal/
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Available at: https://primelawnepal.com/an-overview-of-tax-law-in-nepal/
[Accessed 19 05 2025].

6 Appendix: Logbook Entry Sheet

6.1 Logbook 1

Logbook Entry Sheet

Meeting No: 1 **Date: 1 May**

Start Time: 10 Am **Finish Time: 11 Am**

Items Discussed: coursework was divided among each member.

Achievements: The coursework problem was assigned to all three members and we started working on it.

Problems (if any): Members weren't very cooperative.

Tasks for Next Meeting: To discuss about problem number one.

Student's Name	Student's Signature
AKISHA BHUJEL	Akisha
MANOJ KATWAL	Manoj
BIMAN LIMBU	

6.2 Logbook 2

Logbook Entry Sheet

Meeting No: 2 **Date: 5 May**

Start Time: 10 Pm **Finish Time: 12 Pm**

Items Discussed: To discuss about problem number one.

Achievements: Using excel to solve the tax calculation and creating a spreadsheet that calculate tax on the basic of income and tax rates. The spreadsheet handles range of tax rates, income levels. It helps to analyze tax calculation of monthly income and tax calculation.

Problems (if any): No problem was occurred.

Tasks for Next Meeting: To discuss about problem number two.

Student's Name	Student's Signature
AKISHA BHUJEL	Akisha
MANOJ KATWAL	Manoj
BIMAN LIMBU	

6.3 Logbook 3

Logbook Entry Sheet

Meeting No: 3 **Date: 12 May**

Start Time: 10 Pm **Finish Time: 12 Pm**

Items Discussed: To discuss about problem number two.

Achievements: After solving problem 2 , we now used excel to find minimum and maximum value of part A. Then using demos graph to solve points of part B using graphical method. We have successfully completed problem 2.

Problems (if any): There were many mistakes in part A and the graph plotted was incorrect but later we corrected it.

Tasks for Next Meeting: To discuss about problem number three documentation.

Student's Name	Student's Signature
AKISHA BHUJEL	Akisha
MANOJ KATWAL	Manoj
BIMAN LIMBU	[illegible]

6.4 Logbook 4

Logbook Entry Sheet

Meeting No: 4 **Date: 18 May**

Start Time: 6 Pm **Finish Time: 7 Pm**

Items Discussed: To discuss about problem number three and documentation.

Achievements: We successfully completed problem 3 and documentation was also divided to each member and completed before deadline.

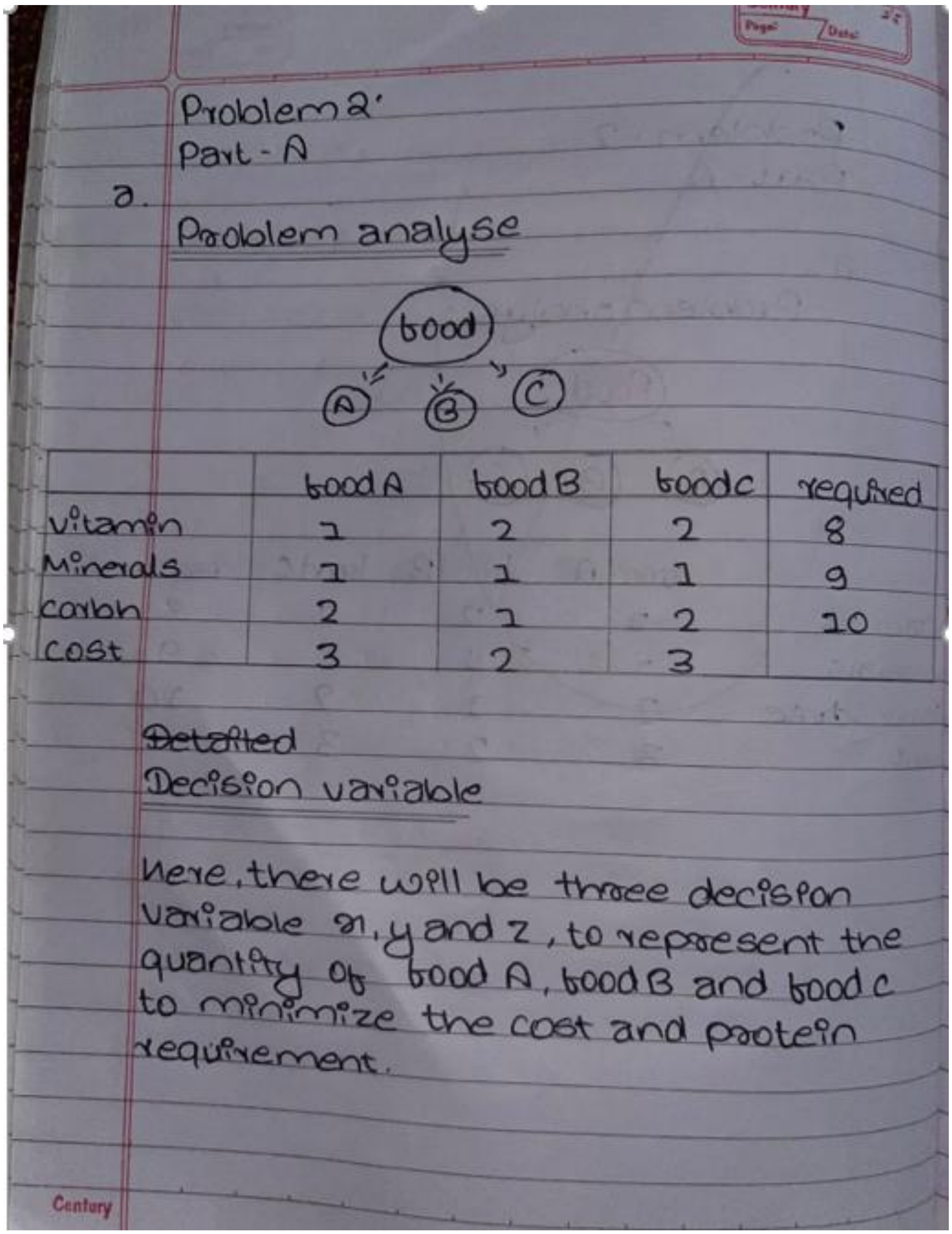
Problems (if any): No problem was occurred.

Tasks for Next Meeting: Coursework was completed no meeting after this.

Student's Name	Student's Signature
AKISHA BHUJEL	Akisha
MANOJ KATWAL	Manoj
BIMAN LIMBU	

7 Evidence

7.1 Problem 2 Part A

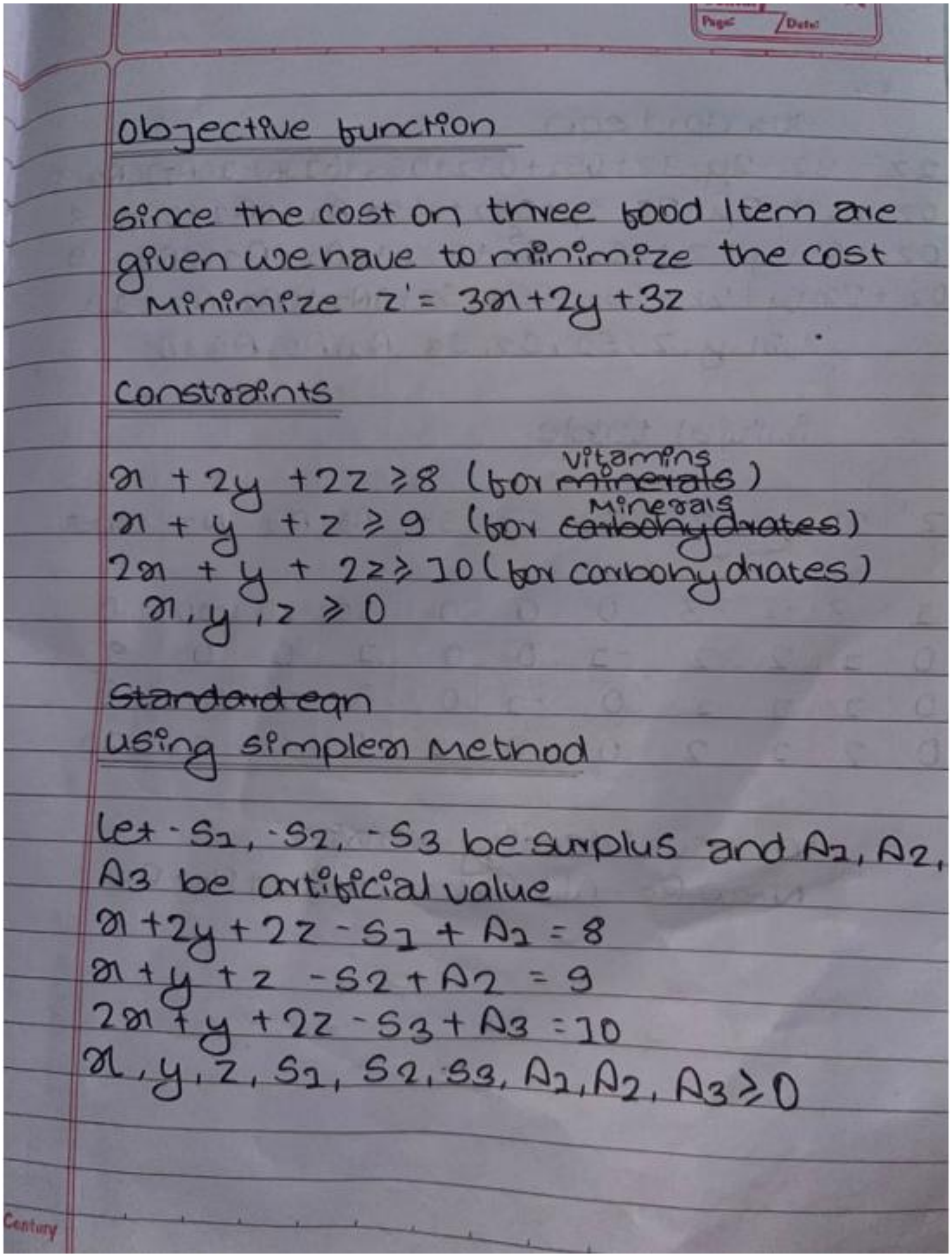

Problem 2'
Part - A
a.
Problem analyse

	food A	food B	food c	required
vitamin	1	2	2	8
Minerals	1	1	1	9
carbh	2	1	2	10
cost	3	2	3	

~~Detailed~~
Decision variable

here, there will be three decision variable x, y and z, to represent the quantity of food A, food B and food c to minimize the cost and protein requirement.

Figure 18:Problem 2 "Part A"(1)



Page: Date:

objective function

since the cost on three food Item are given we have to minimize the cost

Minimize $z' = 3x + 2y + 3z$

Constraints

$x + 2y + 2z \geq 8$ (for vitamins ~~minerals~~)

$x + y + z \geq 9$ (for Minerals ~~carbohydrates~~)

$2x + y + 2z \geq 10$ (for carbohydrates)

$x, y, z \geq 0$

~~Standard eqn~~

using simplex Method

let $-S_1, -S_2, -S_3$ be surplus and A_1, A_2, A_3 be artificial value

$x + 2y + 2z - S_1 + A_1 = 8$

$x + y + z - S_2 + A_2 = 9$

$2x + y + 2z - S_3 + A_3 = 10$

$x, y, z, S_1, S_2, S_3, A_1, A_2, A_3 \geq 0$

Century

Figure 19:Problem 2"Part A" (2)

b.

Standard eqn

$1z' - 3x - 2y - 3z + 0S_1 + 0S_2 + 0S_3 + (-10A) - 10A_2 - 10A_3 = 0$

$0z' + 1x + 2y + 2z - S_1 + 0S_2 + 0S_3 + A_1 + 0A_2 + 0A_3 = 8$

$0z' + x + y + z + 0S_1 - S_2 + 0S_3 + 0A_1 + A_2 + 0A_3 = 9$

$0z' + 2x + y + 2z + 0S_1 + 0S_2 - S_3 + 0A_1 + 0A_2 + A_3 = 10$

$z', x, y, z, S_1, S_2, S_3, A_1, A_2, A_3 \geq 0$

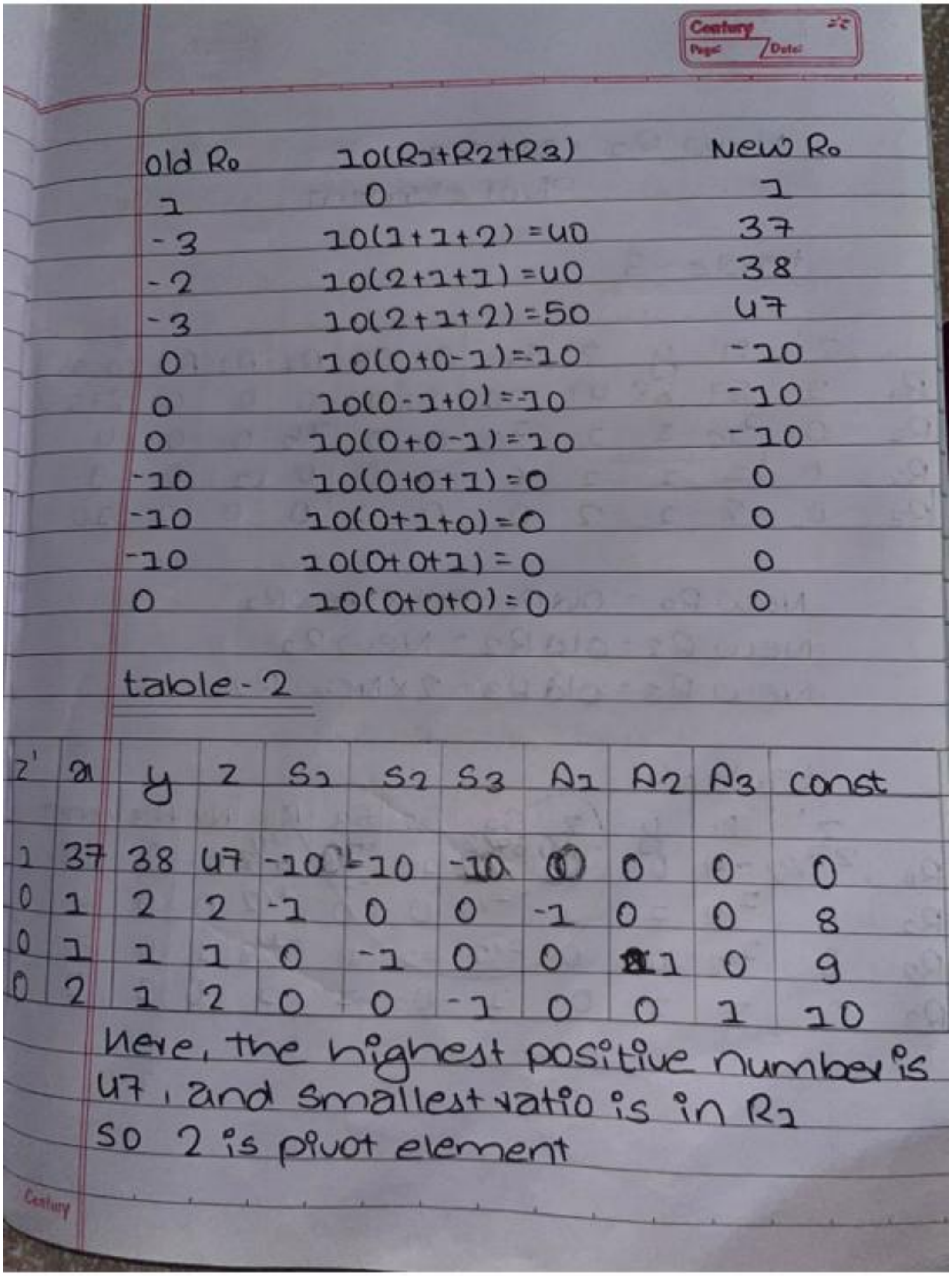
initial table

z'	x	y	z	S_1	S_2	S_3	A_1	A_2	A_3	const
1	-3	-2	-3	0	0	0	-10	-10	-10	0
0	1	2	2	-1	0	0	1	0	0	8
0	1	1	1	0	-1	0	0	1	0	9
0	2	1	2	0	0	-1	0	0	1	10

for new identity Matrix

New R_0 = Old R_0 + $10(R_1 + R_2 + R_3)$

Figure 20:Problem 2"Part A" (3)



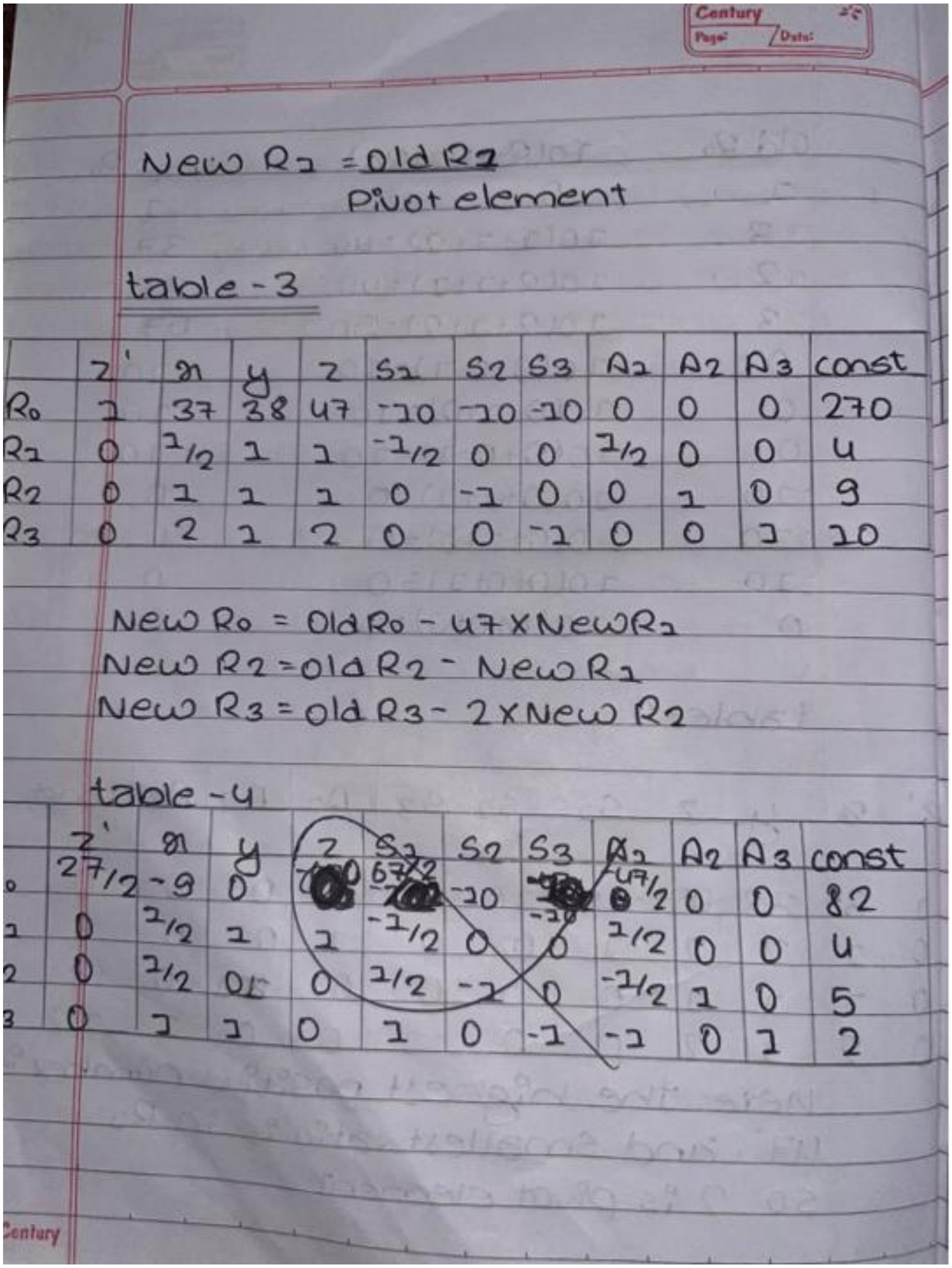
Old R_0	$10(R_1+R_2+R_3)$	New R_0
1	0	1
-3	$10(1+1+2)=40$	37
-2	$10(2+1+1)=40$	38
-3	$10(2+1+2)=50$	47
0	$10(0+0-1)=-10$	-10
0	$10(0-1+0)=-10$	-10
0	$10(0+0-1)=-10$	-10
-10	$10(0+0+1)=0$	0
-10	$10(0+1+0)=0$	0
-10	$10(0+0+1)=0$	0
0	$10(0+0+0)=0$	0

table-2

z'	x	y	z	S_1	S_2	S_3	A_1	A_2	A_3	const
1	37	38	47	-10	-10	-10	0	0	0	0
0	1	2	2	-1	0	0	-1	0	0	8
0	1	1	1	0	-1	0	0	1	0	9
0	2	1	2	0	0	-1	0	0	1	10

here, the highest positive number is 47, and smallest ratio is in R_1 so 2 is pivot element

Figure 21:Problem 2 "Part A" (4)



New R_1 = $\frac{\text{Old } R_1}{\text{Pivot element}}$

table - 3

	z'	x	y	z	S_1	S_2	S_3	A_1	A_2	A_3	const
R_0	1	37	38	47	-10	-10	-10	0	0	0	270
R_1	0	1/2	1	1	-1/2	0	0	1/2	0	0	4
R_2	0	1	1	1	0	-1	0	0	1	0	9
R_3	0	2	1	2	0	0	-1	0	0	1	10

New R_0 = Old R_0 - 47 x New R_1

New R_2 = old R_2 - New R_1

New R_3 = old R_3 - 2 x New R_2

table - 4

	z'	x	y	z	S_1	S_2	S_3	A_1	A_2	A_3	const
0	27/2	-9	0	[illegible]	67/2	-10	[illegible]	47/2	0	0	82
1	0	1/2	1	1	-1/2	0	0	1/2	0	0	4
2	0	1/2	0	0	1/2	-1	0	-1/2	1	0	5
3	0	1	1	0	1	0	-1	-1	0	1	2

Figure 22:Problem 2 "Part A" (5)

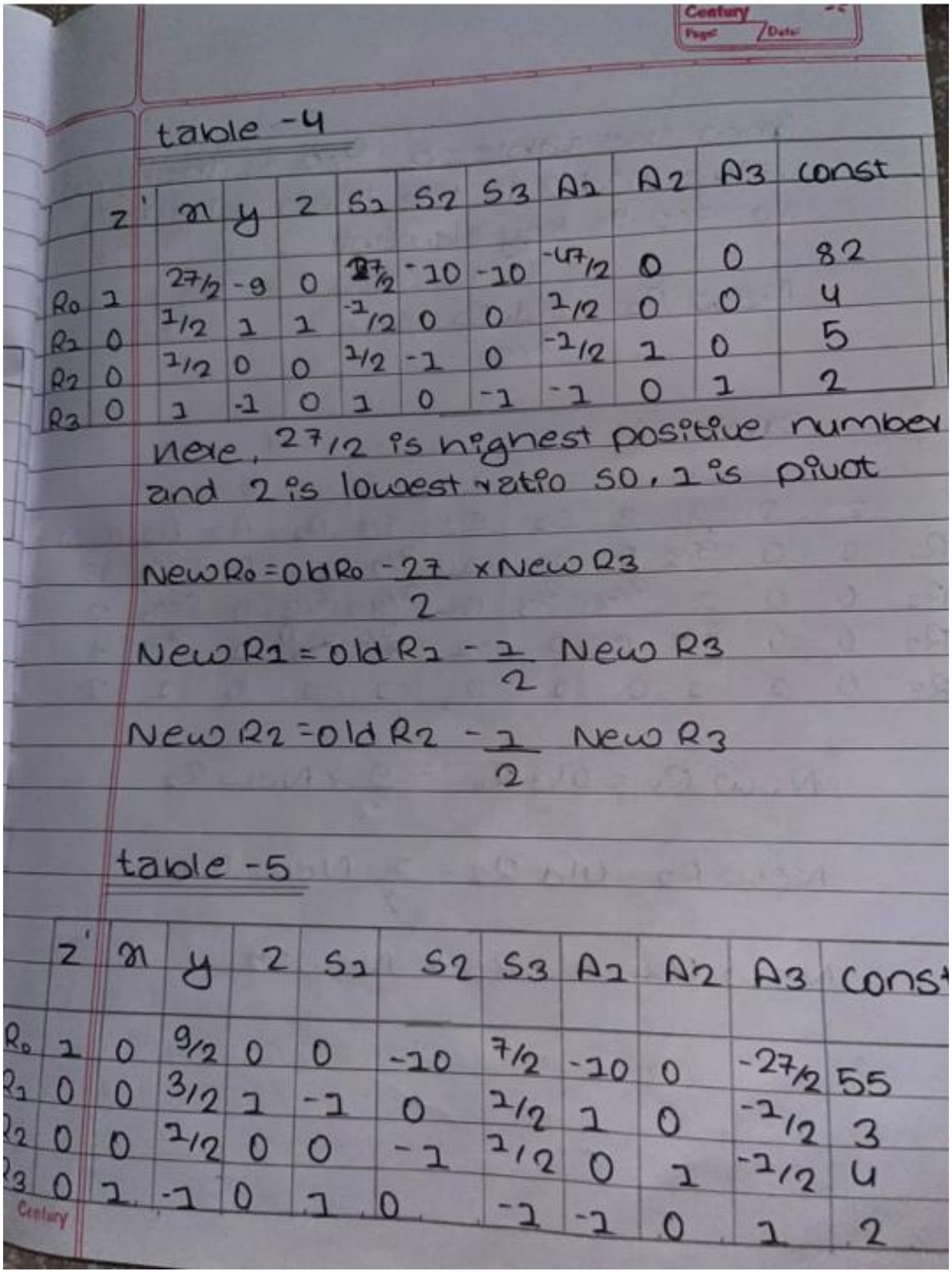


table -4

	z'	x	y	z	S_1	S_2	S_3	A_1	A_2	A_3	const
R_0	1	27/2	-9	0	27/2	-10	-10	-47/2	0	0	82
R_1	0	1/2	1	1	-1/2	0	0	1/2	0	0	4
R_2	0	1/2	0	0	1/2	-1	0	-1/2	1	0	5
R_3	0	1	-1	0	1	0	-1	-1	0	1	2

here, 27/2 is highest positive number and 2 is lowest ratio so, 1 is pivot

$New R_0 = old R_0 - \frac{27}{2} \times New R_3$

$New R_1 = old R_1 - \frac{1}{2} New R_3$

$New R_2 = old R_2 - \frac{1}{2} New R_3$

table -5

	z'	x	y	z	S_1	S_2	S_3	A_1	A_2	A_3	const
R_0	1	0	9/2	0	0	-10	7/2	-10	0	-27/2	55
R_1	0	0	3/2	1	-1	0	1/2	1	0	-1/2	3
R_2	0	0	1/2	0	0	-1	1/2	0	1	-1/2	4
R_3	0	1	-1	0	1	0	-1	-1	0	1	2

Figure 23:Problem 2 "Part A" (6)

from the table -5 $9/2$ is the higest value and 2 is the lower value ratio so $3/2$ is key element

$$\text{New } R_1 = \frac{\text{Old } R_1}{3/2}$$

table - 6

	z'	x	y	z	S_1	S_2	S_3	A_1	A_2	A_3	const
R_0	1	0	$9/2$	0	0	-10	$7/2$	-10	0	$-27/2$	55
R_1	0	0	1	$2/3$	$-2/3$	0	$1/3$	$2/3$	0	$-1/3$	2
R_2	0	0	$1/2$	0	0	-1	$\cancel{8}\,7/2$	$\cancel{1\times2}\,0$	1	$-1/2$	4
R_3	0	1	-1	0	1	0	-1	-1	0	1	2

$$\text{New } R_0 = \text{Old } R_0 - \frac{9}{2} \times \text{New } R_1$$

$$\text{New } R_2 = \text{Old } R_2 - \frac{1}{2} \text{ New } R_1$$

$$\text{New } R_3 = \text{Old } R_3 + \text{New } R_1$$

Figure 24:Problem 2 "Part A" (7)

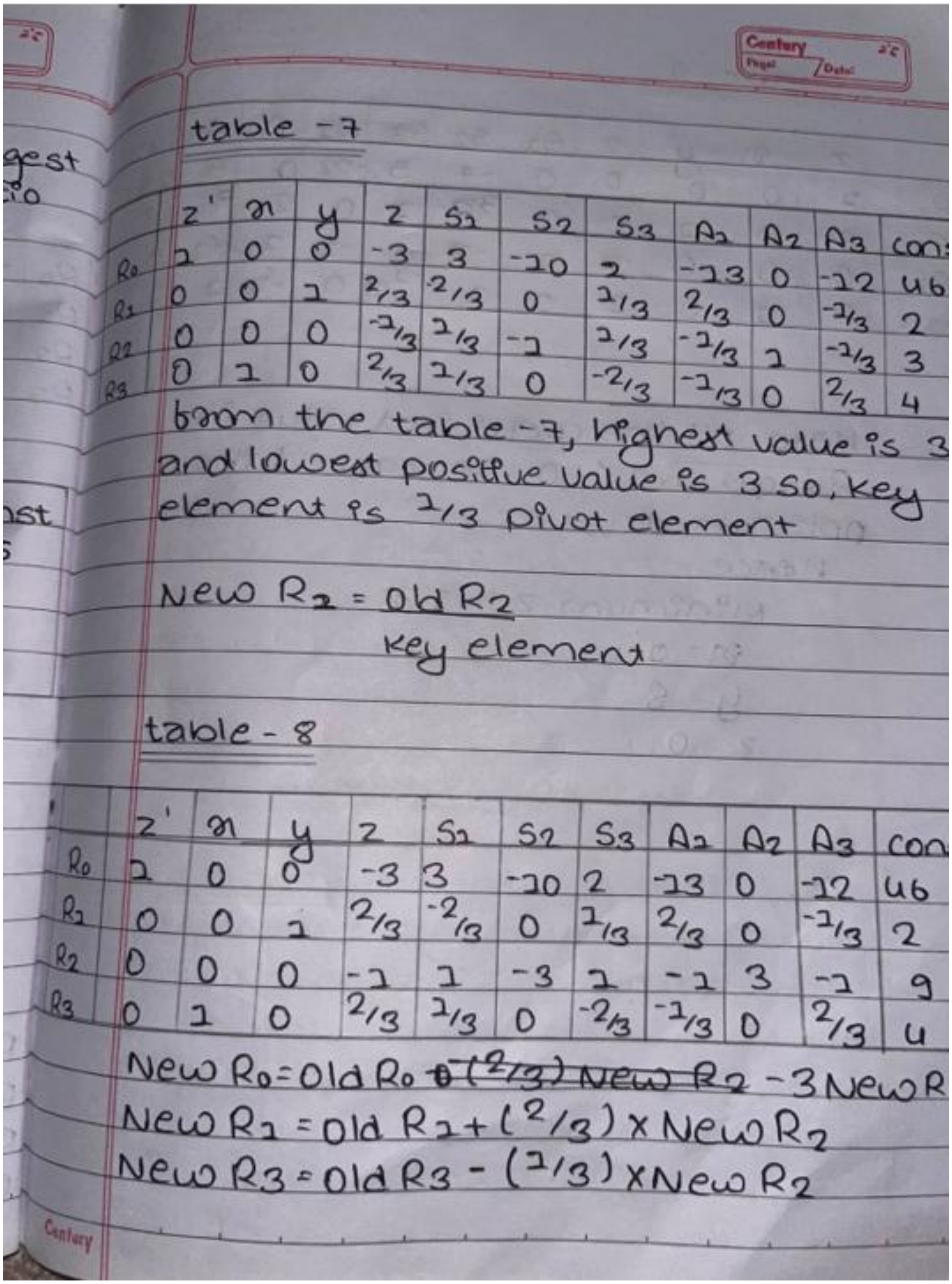


table - 7

	z'	x	y	z	S_1	S_2	S_3	A_1	A_2	A_3	con
R_0	1	0	0	-3	3	-10	2	-13	0	-12	46
R_1	0	0	1	2/3	-2/3	0	1/3	2/3	0	-1/3	2
R_2	0	0	0	-1/3	1/3	-1	1/3	-1/3	1	-1/3	3
R_3	0	1	0	2/3	1/3	0	-2/3	-1/3	0	2/3	4

from the table - 7, highest value is 3 and lowest positive value is 3 so, key element is 1/3 pivot element

New $R_2 = \frac{\text{Old } R_2}{\text{key element}}$

table - 8

	z'	x	y	z	S_1	S_2	S_3	A_1	A_2	A_3	con
R_0	1	0	0	-3	3	-10	2	-13	0	-12	46
R_1	0	0	1	2/3	-2/3	0	1/3	2/3	0	-1/3	2
R_2	0	0	0	-1	1	-3	1	-1	3	-1	9
R_3	0	1	0	2/3	1/3	0	-2/3	-1/3	0	2/3	4

New R_0 = Old R_0 ~~+ (2/3) New R_2~~ - 3 New R

New R_1 = Old R_1 + (2/3) x New R_2

New R_3 = Old R_3 - (1/3) x New R_2

Figure 25:Problem 2"Part A" (8)

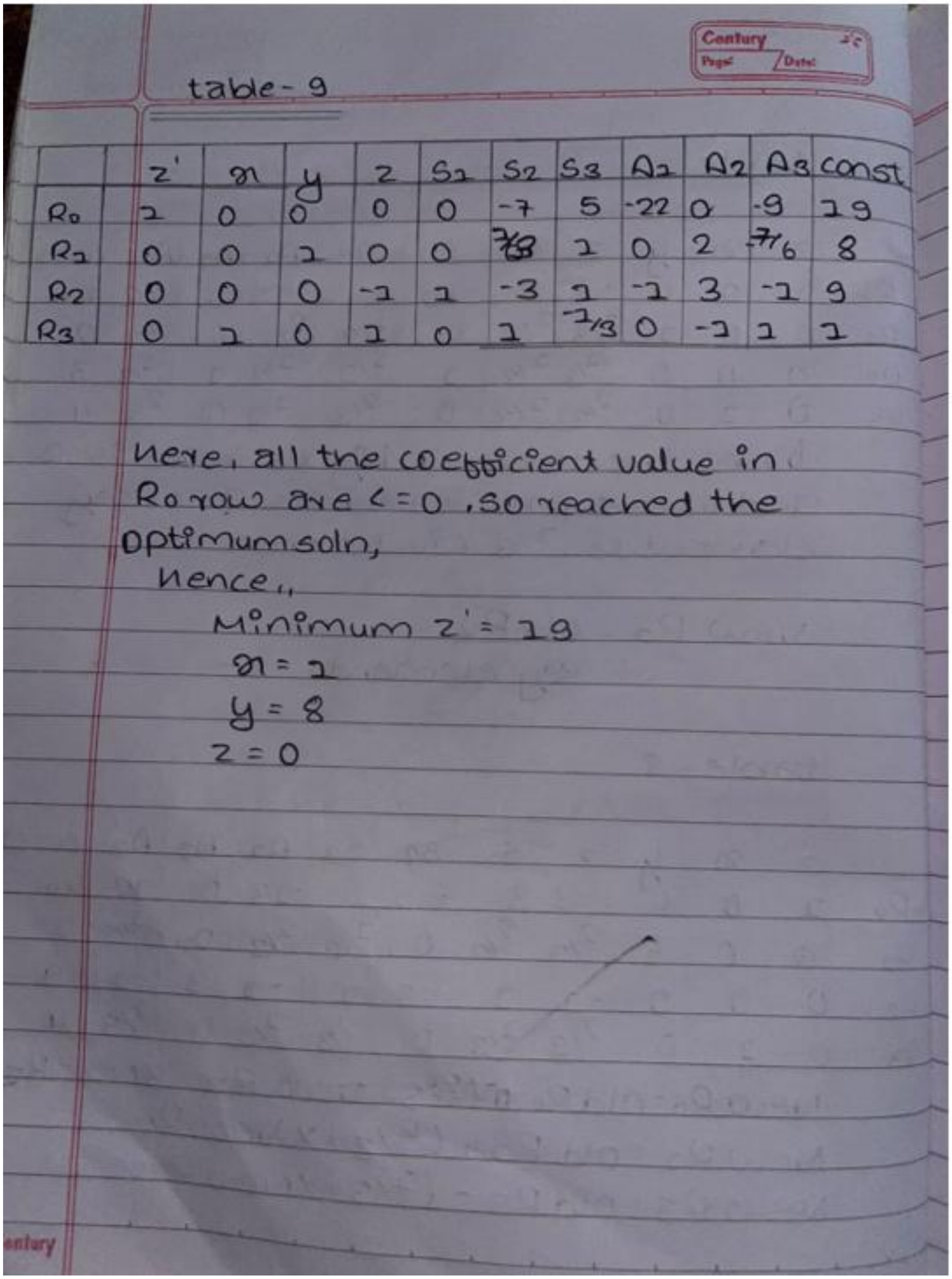


table- 9

	z'	x	y	z	S_1	S_2	S_3	A_1	A_2	A_3	const
R_0	1	0	0	0	0	-7	5	-22	0	-9	19
R_1	0	0	1	0	0	7/3	1	0	2	-7/6	8
R_2	0	0	0	-1	1	-3	1	-1	3	-1	9
R_3	0	1	0	1	0	1	-1/3	0	-1	1	1

here, all the coefficient value in R_0 row are <=0 ,so reached the optimum soln,

hence,,

Minimum $z' = 19$

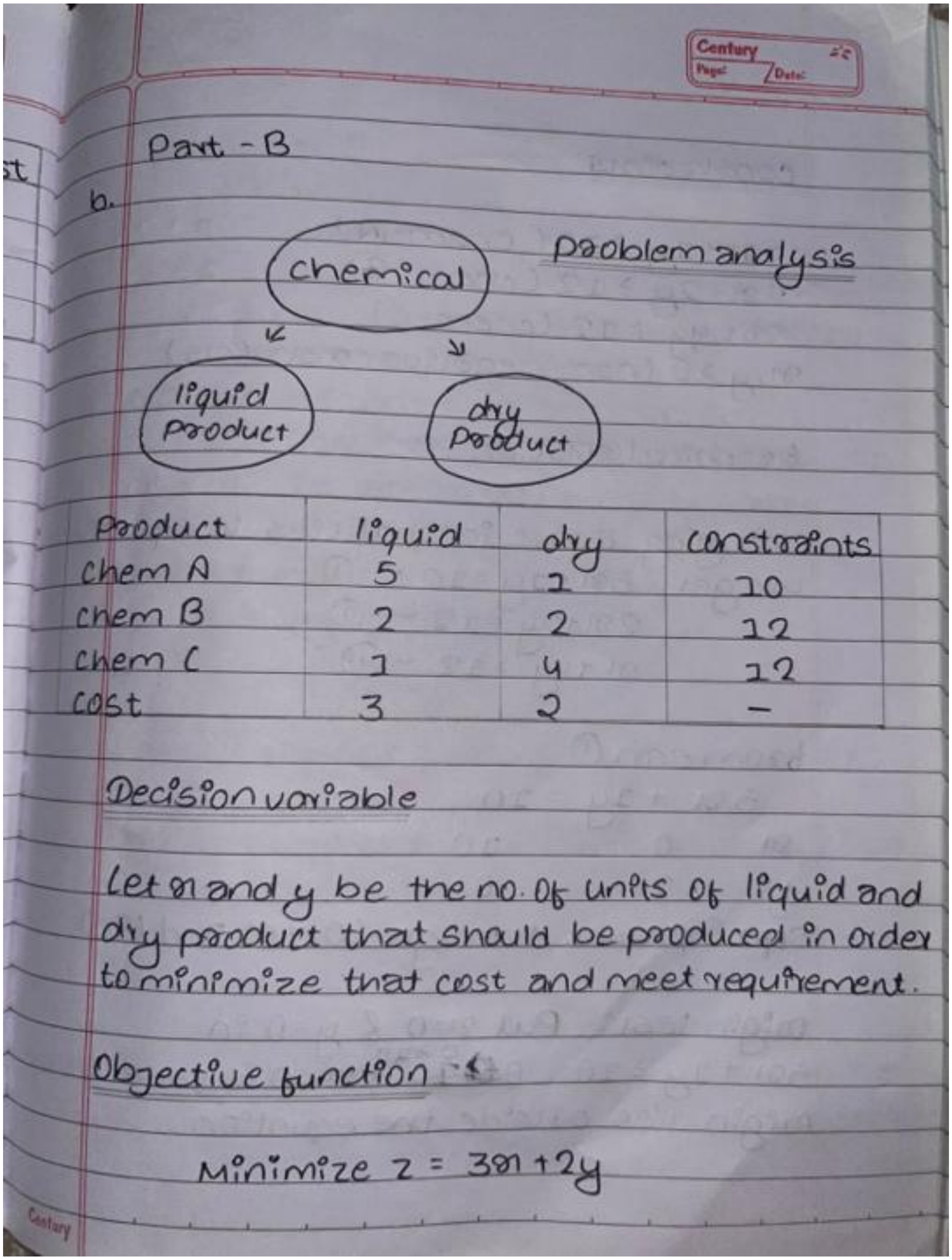
$x = 1$

$y = 8$

$z = 0$

Figure 26:Problem 2 "Part A" (9)

7.2 Problem 2 part 2



Part - B

b.

problem analysis

chemical

liquid product

dry product

Product	liquid	dry	constraints
chem A	5	1	10
chem B	2	2	12
chem C	1	4	12
cost	3	2	–

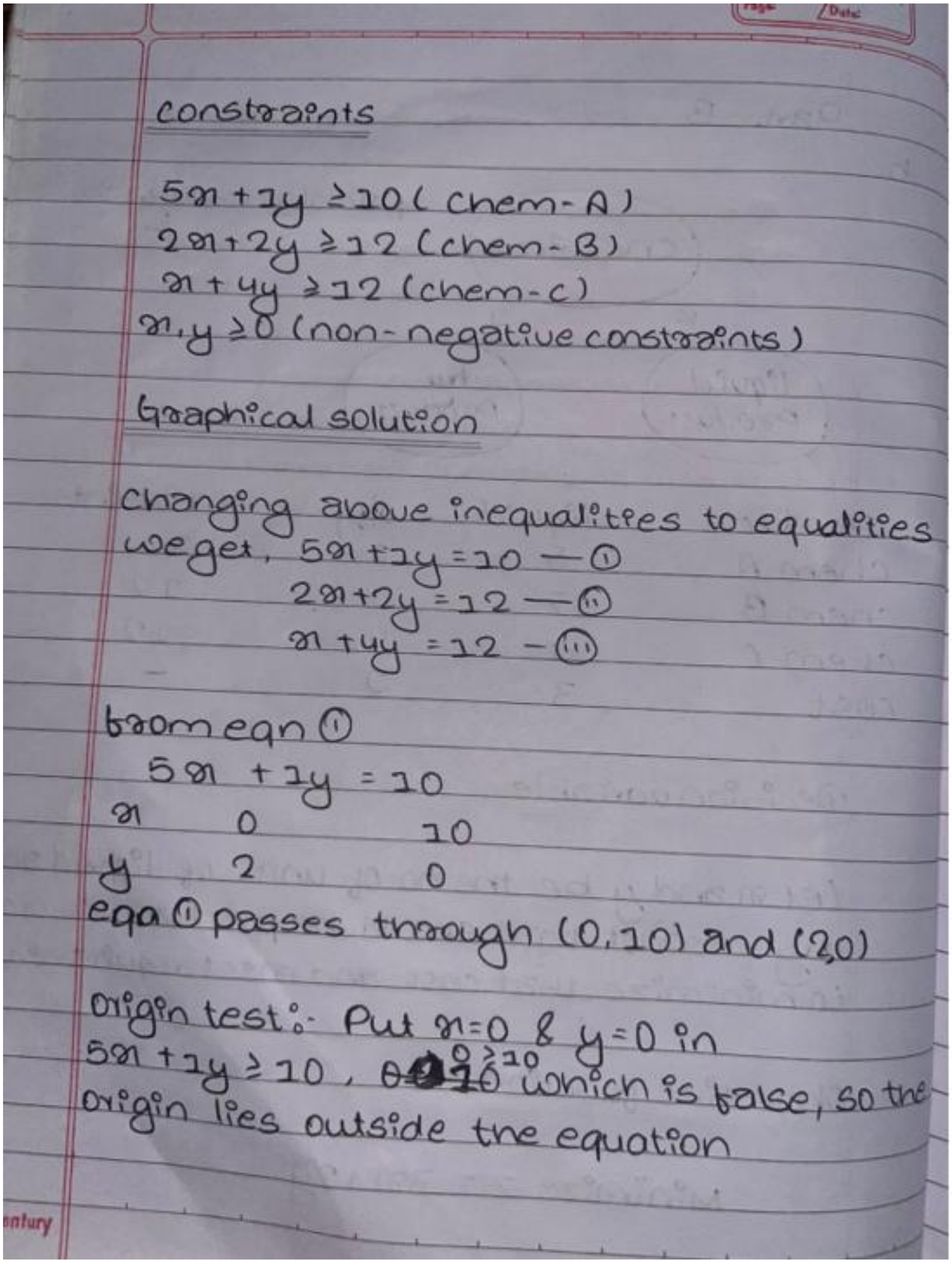
Decision variable

Let x and y be the no. of units of liquid and dry product that should be produced in order to minimize that cost and meet requirement.

Objective function :-

Minimize $z = 3x + 2y$

Figure 27:Problem 2 "Part B" (1)



Constraints

$5x + 1y \geq 10$ (chem-A)
$2x + 2y \geq 12$ (chem-B)
$x + 4y \geq 12$ (chem-C)
$x, y \geq 0$ (non-negative constraints)

Graphical solution

Changing above inequalities to equalities we get, $5x + 1y = 10$ — (i)
$2x + 2y = 12$ — (ii)
$x + 4y = 12$ — (iii)

from eqn (i)
$5x + 1y = 10$

x	0	10
y	2	0

eqn (i) passes through (0,10) and (2,0)

Origin test:- Put $x=0$ & $y=0$ in
$5x + 1y \geq 10$, $0 \geq 10$ which is false, so the origin lies outside the equation

Figure 28:Problem 2 "Part B" (2)

from eqn (ii)

$$2x + 2y = 12$$

x	0	6
y	6	0

eqn (ii) passes through (0,6) and (6,0)

Origin test:- put $x=0$ and $y=0$ in $2x+2y \geq 12$, $0 \geq 12$ which is false, so the origin lies outside the equation

from eqn (iii)

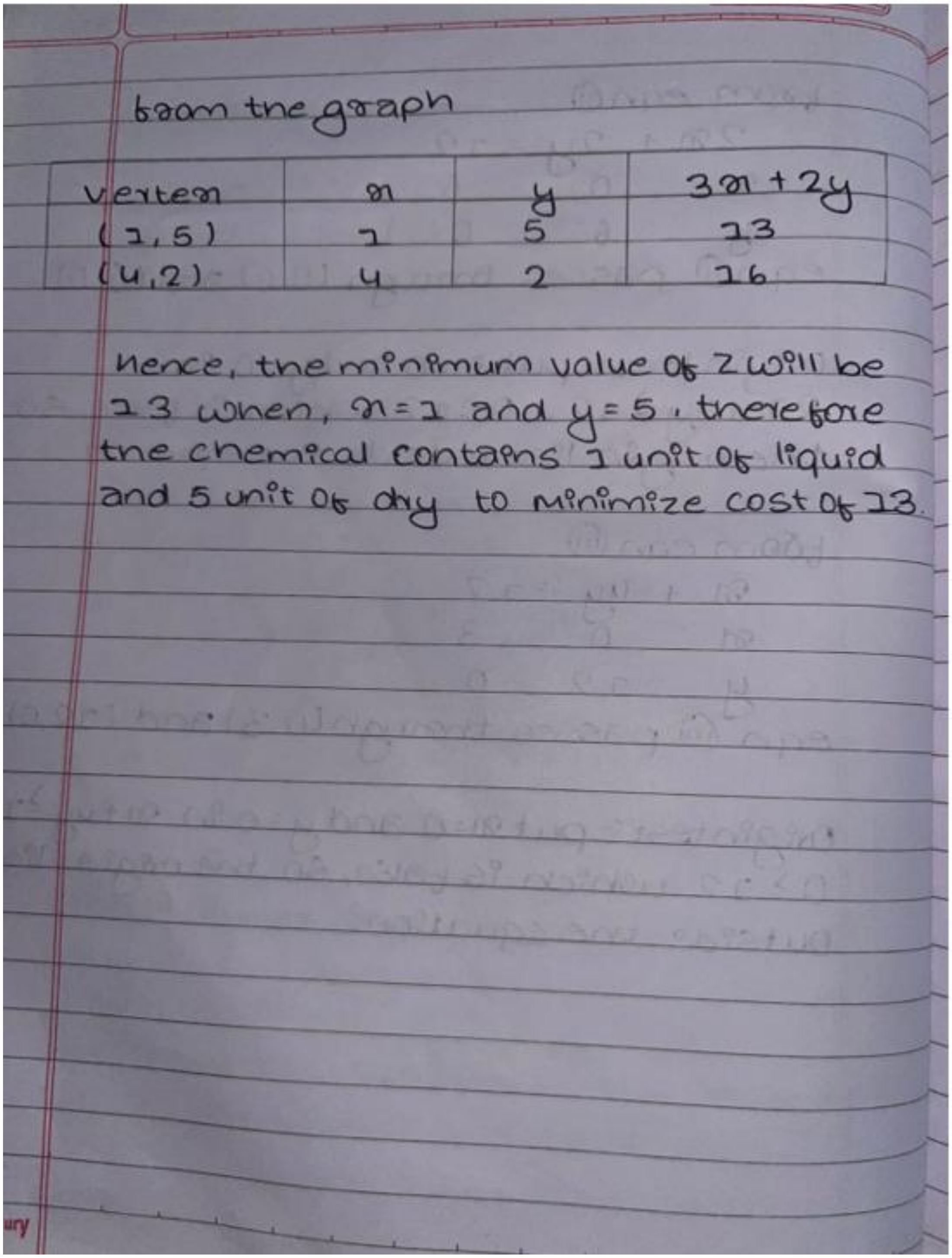
$$x + 4y = 12$$

x	0	3
y	12	0

eqn (iii) passes through (0,3) and (12,0)

Origin test:- put $x=0$ and $y=0$ in $x+4y \geq 12$
$0 \geq 12$ which is false, so the origin lies outside the equation.

Figure 29:Problem 2 "Part B" (3)



from the graph

vertex	x	y	$3x + 2y$
(1, 5)	1	5	13
(4, 2)	4	2	16

hence, the minimum value of Z will be 13 when, $x = 1$ and $y = 5$. therefore the chemical contains 1 unit of liquid and 5 unit of dry to minimize cost of 13.

Figure 30:Problem 2 :Part B'' (4)

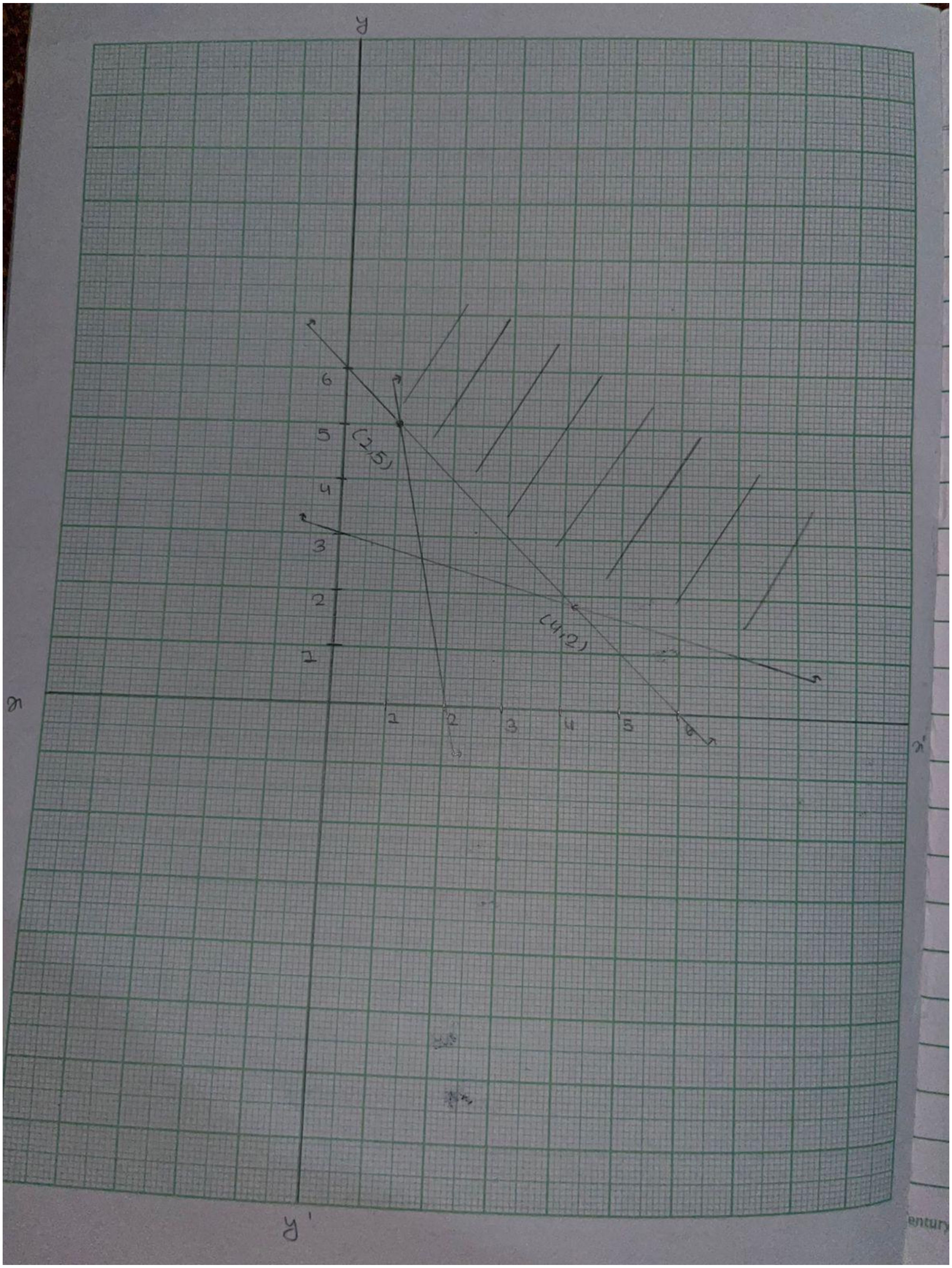
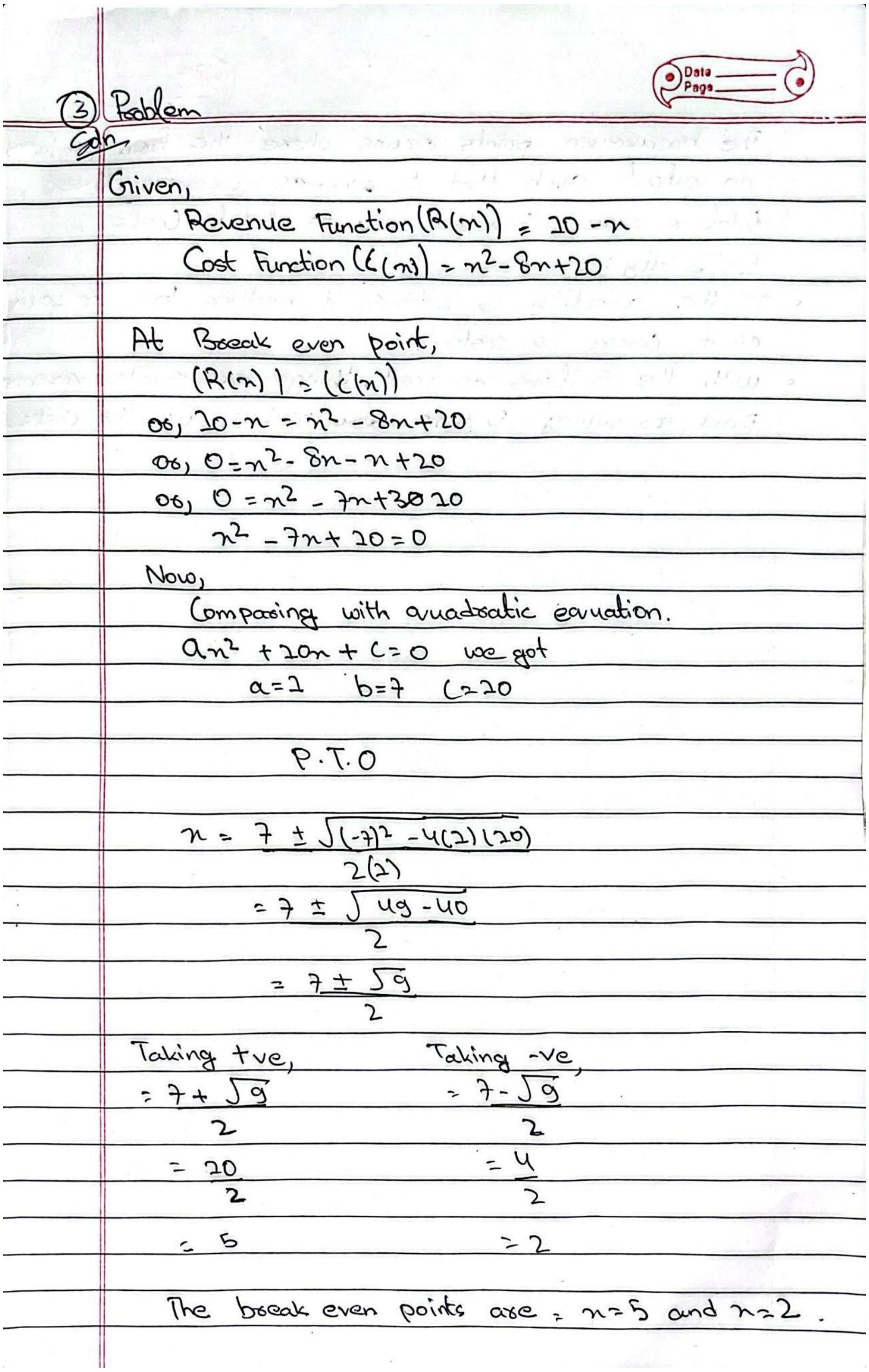


Figure 31:Problem 2 "Part B" (5)

7.3 Problem 3



③ Problem

Soln

Given,

Revenue Function $(R(x)) = 10 - x$

Cost Function $(C(x)) = x^2 - 8x + 20$

At Break even point,

$(R(x)) = (C(x))$

or, $10 - x = x^2 - 8x + 20$

or, $0 = x^2 - 8x - x + 20$

or, $0 = x^2 - 7x + \cancel{30}\,10$

$x^2 - 7x + 10 = 0$

Now,

Comparing with quadratic equation.

$ax^2 + 10x + c = 0$ we got

$a = 1 \quad b = 7 \quad c = 10$

P.T.O

$$x = \frac{7 \pm \sqrt{(-7)^2 - 4(1)(10)}}{2(1)}$$

$$= \frac{7 \pm \sqrt{49 - 40}}{2}$$

$$= \frac{7 \pm \sqrt{9}}{2}$$

Taking +ve,

$$= \frac{7 + \sqrt{9}}{2}$$

$$= \frac{10}{2}$$

$$= 5$$

Taking -ve,

$$= \frac{7 - \sqrt{9}}{2}$$

$$= \frac{4}{2}$$

$$= 2$$

The break even points are $x = 5$ and $x = 2$.

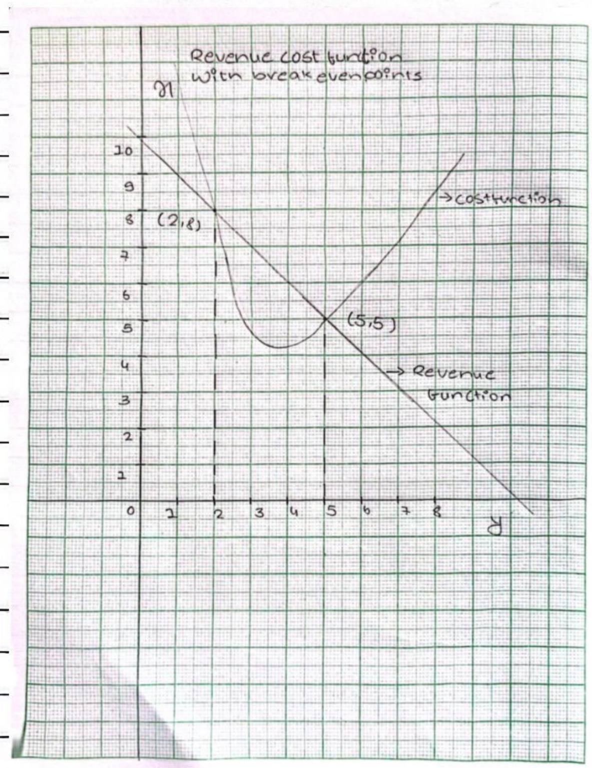
Figure 32:Problem 3 (1)

Date
Page

The breakeven points occurs where the firm produces an output such that it ~~receives~~ recovers the total revenue collected and its total costs.

Especially:

- If the quantity is 2 items, it neither loss or gain as it covers its costs.
- With the 5 items as well, things are similar revenue paid for selling 5 items also makes up the costs.



Output (x)

Here, the graph showing the revenue and cost functions, with the break even points clearly at (2,8) and (5,5)

Figure 33:Problem 2 (2)

(a)
Soln

i) Profit Function,

$$P(x) = R(x) - C(x)$$
$$= (20x) - (x^2 - 8x + 20)$$
$$= 20x - x - x^2 + 8x - 20$$
$$= -x^2 + 7x - 20$$

~~Comparing with the quadratic equation that opens downward (x^2 is negative)~~

$$x = \frac{-b}{2a}$$

where $a = -1$ and $b = 7$

$$x = \frac{-7}{2(-1)}$$
$$= \frac{7}{2}$$
$$= 3.5 \text{ units}$$

ii) Maximum Profit,

$$= c - \frac{b^2}{4a}$$
$$= -20 - \frac{(7)^2}{4 \times -1}$$
$$= \frac{-20 \times -4 - 49}{-4}$$
$$= \frac{+9}{+4}$$
$$= \$2.25$$

Therefore, the level production that maximizes the profit is 3.5 units and maximum Profits is \$2.25.

Figure 34:Problem 3 (3)